Conformal Bernstein–Bézier Splines on Compact Two-Manifolds of Positive Genus: Construction, Algorithm, and Experiments

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Abstract

Every smooth closed two-dimensional Riemannian manifold is isometric to a quotient of one of the three constant-curvature model surfaces by a group of isometries, carrying a conformal metric: the round sphere in genus zero, the flat plane in genus one, and the hyperbolic plane in every higher genus, with the nonorientable surfaces arising from quotients by groups containing orientation-reversing isometries. This paper treats the genus one and higher cases with smooth splines in broken Bernstein–Bézier form, continuing the genus-zero construction. Everything is two-dimensional: the surfaces are compact two-manifolds, closed or with a boundary of closed model geodesics, orientable or not, the model surfaces are two-dimensional, and the discretization is by bivariate splines on a planar or hyperbolic polygon.

Topology enters through the identification of the sides of a fundamental polygon, which appears in the computation only as additional rows of the smoothness matrix; geometry enters through the conformal factor, a single positive function on the polygon that multiplies the load vector, the mean-value constraint and the mass matrix. The stiffness matrix is the flat Dirichlet matrix of the polygon for every surface.

The construction is carried out for the torus, the Klein bottle, the Möbius strip and the Bolza surface, with the Gauss–Bonnet theorem used at every level as the check on the identifications, and the Poisson equation and the Laplace–Beltrami eigenvalue problem are solved on them. The experiments reproduce the closed-form spectra of the flat quotients, the rates $d+1$ in L^2 and d in energy for the Poisson problem on a conformal torus and a conformal Klein bottle, the rate $2d$ for the eigenvalues of a conformal Möbius band against a Fourier reference and of the Bolza surface against published values, and measurements of the conditioning and sparsity of the constrained systems.

Keywords. Laplace–Beltrami operator, compact two-manifolds, spline quotients, fundamental polygon, side pairings, conformal factor, bivariate Bernstein–Bézier methods, Gauss–Bonnet theorem

1 Introduction

Throughout this paper a surface means a smooth compact two-dimensional Riemannian manifold, closed or with a boundary of the kind admitted in Section 6.1, with or without an embedding in $\mathbb{R}^3$ and orientable or not, and every construction is two-dimensional: the domains are planar or hyperbolic polygons, the splines are bivariate, and the differential operator is the Laplace–Beltrami operator of a two-dimensional metric. That operator is the model problem for second-order elliptic computation on surfaces, and arises in geometry processing, in the modeling of diffusion on interfaces and membranes, and in spectral shape analysis [20, 9, 37]. The genus-zero paper [25] discretizes it by writing the surface as the unit sphere with a conformal metric $e^{2\lambda} g_{\mathbb{S}^2}$, so that the whole geometry is one scalar function λ on the sphere. Two identities carry the computation: the area element of

the conformal metric is $e^{2\lambda}$ times the round one, and the Dirichlet energy is the round energy. The stiffness matrix is therefore the round-sphere matrix for every surface and the surface enters only through $w = e^{2\lambda}$ at the quadrature points of the load vector, the mean-value constraint and the weighted mass matrix.

This paper continues the construction to every other closed surface. By the uniformization theorem [23, Ch. 2], a smooth closed Riemannian surface M is isometric to

$$M \;=\; \big(\, X/\Gamma,\; e^{2\lambda} g_X \,\big), \tag{1}$$

where $\chi(M) = V - E + T$ is the Euler characteristic of any triangulation of M with V vertices, E edges and T triangles, X is the flat plane $\mathbb{E}^2$ when $\chi(M) = 0$ and the hyperbolic plane $\mathbb{H}^2$ when $\chi(M) < 0$, Γ is a discrete group of isometries of X acting freely on X, and λ is a smooth function on X invariant under Γ. For a closed orientable surface of genus g one has $\chi = 2 - 2g$, and a closed nonorientable surface with k cross-caps has $\chi = 2 - k$; positive genus in the title means $\chi \le 0$. For a compact surface with boundary the same description holds with X replaced by a Γ-invariant closed region $\Omega \subseteq X$ bounded by geodesics, $M = \Omega/\Gamma$; the Möbius band is Ω/Γ with $\Omega = \mathbb{R} \times [0, b]$ and Γ generated by a glide reflection along the strip. Throughout, X/Γ is written for Ω/Γ when the distinction is immaterial, and $\Omega = X$ for a closed surface. The quotient is presented by a fundamental polygon $F \subset \Omega$ whose sides are identified in pairs by elements of Γ, the free sides, if any, lying on $\partial\Omega$; this is the orbifold picture of a closed surface, used for spline modeling by Wallner and Pottmann [41], who observed that all three models carry Bernstein–Bézier polynomials because their isometry groups act linearly, and continued in the rational orbifold splines of Beccari and Prautzsch [6]: a square for the torus and the Klein bottle, a strip for the Möbius strip, an octagon for the Bolza surface. This gives the two channels of the title. Topology enters through the side identifications of F, and in the constrained Bernstein–Bézier form of the spline space these are additional rows of the smoothness matrix, of the same kind as the rows across an interior edge. Geometry enters through λ, and here the dimension is decisive: in dimension two, and only there, the Dirichlet energy is invariant under a conformal change of metric, so the stiffness matrix is the Dirichlet matrix of the polygon F in the model metric g_X for every λ, and λ enters where the area element does. When $X = \mathbb{H}^2$ the model metric itself is conformal to the flat metric of a disk chart, so even the hyperbolic stiffness matrix is a flat Dirichlet matrix, and the hyperbolic area density joins $e^{2\lambda}$ in the weight.

The contributions are the following.

A single discretization for all closed surfaces of positive genus, orientable or not. The side pairings of the fundamental polygon enter the computation only as additional rows of the smoothness matrix, of the same form (5) used across an interior edge, because an isometry of the model acts trivially on Bernstein coefficients (Proposition 4.1); the resulting coefficient arrays are in bijection with splines on the quotient (Theorem 4.2). Nothing in the statement or the proof distinguishes orientation-preserving from orientation-reversing pairings, so the torus and the Klein bottle, and the Bolza surface and the nonorientable N_3, are the same computation with different rows.

A domain carried without approximation, and a metric-independent stiffness matrix. The polygon and its identifications carry no discretization error at any refinement level (Section 4.3), so no polygonal or isoparametric surface is introduced and the error analysis carries no geometric consistency term. The conformal factor enters the load vector, the mean-value constraint and the mass matrix, and nowhere else; the stiffness matrix is the model Dirichlet matrix of the polygon, assembled once per mesh and degree and reused for every metric on that quotient (Section 6.2).

Error estimates with the hypotheses that quotients require. Theorem 7.1 gives the approximation property of the quotient spline space, proved by transporting a stable local determining set through

the identification of Theorem 4.2, and Corollary 7.2 the rates d in energy, $d+1$ in L^2 and $2d$ for eigenvalues, with constants depending on the conformal factor only through $w_{\min}$, $w_{\max}$ and $\|w\|_{C^1}$, for a fixed quotient. Proposition 6.2 adds three facts that hold at every mesh and degree: one-sidedness of the eigenvalues of the Galerkin pencil, reproduction of the zero mode to roundoff when the constants lie in the space, and a bound showing that replacing the weight by an approximate factor perturbs the whole spectrum by a relative amount uniform in the eigenvalue index.

Checks against closed-form and independent references. The flat quotients are tested against their closed-form spectra, the conformal ones against Fourier–Galerkin references on the double cover, the hyperbolic ones against the published Bolza spectrum, and the identifications themselves against the dimension identities and the Gauss–Bonnet theorem at four levels; a comparison with parametric surface finite elements at equal degree isolates the treatment of the geometry as the one difference.

The relation to the author's companion preprints is as follows. The broken representation with smoothness as constraints and the rank-revealing elimination are taken from [24]; the genus-zero case, in which Γ is trivial and the model is the sphere, is [25]; [26] constructs C^1 spline surfaces on the flat quotients for geometric functionals and does not treat the Laplace–Beltrami operator or conformal metrics; and [27] constructs the hyperbolic spline spaces on side-paired polygons and computes the Bolza spectrum. What is new here is the unified quotient formulation across the flat and hyperbolic models, orientable and not (Proposition 4.1, Theorem 4.2), the intrinsic presentation without a polygon or a group (Section 4.4), the conformal-metric separation of geometry from topology (Lemma 5.1) with the error analysis of Section 7 that rests on it, the genus ladder, the equal-degree comparison with surface finite elements, and every experiment on a conformal metric; the Bolza reference computation of Table 10 is repeated here in the present framework and agrees with [27]. All four companion manuscripts are preprints under review at the time of writing, and none is a prerequisite for this one.

Two matters are outside the scope of this paper. For a surface given only as a triangle mesh the pair (Γ, λ) must be computed by discrete uniformization, which is the subject of a companion paper; and the dependence of the constants in Corollary 7.2 on the conformal factor and on the position in the spectrum is not made explicit here.

2 Related work

Splines on the constant-curvature models. Spherical Bernstein–Bézier polynomials and spherical splines are due to Alfeld, Neamtu and Schumaker [3, 2], with the approximation theory of Neamtu and Schumaker [35] and [32, Ch. 13–14]. That the same construction runs on all three Cayley–Klein models, because their isometry groups act linearly, was observed by Wallner and Pottmann [41], who also described fundamental polygons with Poincaré relations and the nonorientable words $a_1^2 \cdots a_g^2$; what that paper does not contain is the constraint machinery over a fixed triangulation, dimension results, an approximation theory or numerics, and the present work realizes as numerical analysis the programme it announced. The broken Bernstein–Bézier representation with smoothness as linear constraints, and the rank-revealing elimination used here, are from [24]; the genus-zero case of the present construction is [25], the hyperbolic case [27], and the flat case with immersed spline surfaces [26].

Finite elements on surfaces. The standard alternative is the surface finite element method of Dziuk [19], surveyed in [20] and, with a full account of the geometric error, in Bonito, Demlow and Nochetto [9]; higher-order parametric elements and their estimates are due to Demlow [16], and the a priori theory for the Laplace–Beltrami eigenvalue problem to Bonito, Demlow and Owen [10].

In all of these the surface is approximated and a geometric consistency error enters alongside the Galerkin error. The construction here removes that term for the surfaces it covers, at the price of requiring the uniformizing data as input.

Spectra as references. The Bolza eigenvalues computed to high precision by Strohmaier and Uski [39] are the external reference for the hyperbolic experiments, with the multiplicities established in [22, 15]; the Klein quartic, whose first eigenvalue has multiplicity eight [11], appears here only as a second example of a side-paired polygon. The flat quotients have elementary closed-form spectra. The Galerkin approximation theory of symmetric eigenvalue problems used throughout is that of Babuška and Osborn [4] and Boffi [7].

3 Bernstein–Bézier splines on the model surfaces

The spline layer is the one of the genus-zero paper with the sphere replaced by the plane or the hyperboloid. We recall only what the quotient construction uses.

3.1 The three models and their barycentric coordinates

Let $Q_\kappa = \operatorname{diag}(\kappa, 1, 1)$ and $\langle x, y\rangle_\kappa = x^\top Q_\kappa y$ on $\mathbb{R}^3$. The three model surfaces are the quadrics $X_\kappa = \{x : \langle x, x\rangle_\kappa = \kappa,\ x_0 > 0\}$: the unit sphere for $\kappa = 1$, the upper sheet of the hyperboloid $-x_0^2 + x_1^2 + x_2^2 = -1$ for $\kappa = -1$, and, for $\kappa = 0$, the plane $x_0 = 1$, which carries the flat metric. Their isometry groups act linearly on $\mathbb{R}^3$, by $O(3)$, $O^+(2,1)$ and the affine group of the plane written as 3×3 matrices fixing x_0, and this is the reason the Bernstein–Bézier apparatus survives on all three: a linear map carries homogeneous polynomials to homogeneous polynomials.

For a geodesic triangle $T = \langle v_1, v_2, v_3\rangle \subset X_\kappa$ with $V = [v_1\, v_2\, v_3]$ nonsingular, the barycentric coordinates of $x \in X_\kappa$ are the solution $b(x)$ of $Vb = x$, linear in x, and the Bernstein basis of degree d is

$$B^d_{ijk}(x) = \frac{d!}{i!\, j!\, k!}\, b_1(x)^i\, b_2(x)^j\, b_3(x)^k, \qquad i + j + k = d, \tag{2}$$

the restriction to X_κ of a homogeneous trivariate polynomial. For $\kappa = 0$ the coordinates satisfy $b_1 + b_2 + b_3 = 1$ and (2) is the classical planar basis. For $\kappa \neq 0$ the sum $b_1 + b_2 + b_3$ is not 1, the space $\mathcal{B}_d$ spanned by (2) contains the constants if and only if d is even, and this parity restriction governs the choice of degree for scalar problems on the sphere and the hyperboloid [25, 27]. On the plane every degree contains the constants.

3.2 Domain points, B-coefficients and smoothness conditions

We use the notation of Lai and Schumaker [32] throughout. For a triangle $T = \langle v_1, v_2, v_3\rangle$ of X_κ the domain points of degree d are

$$\mathcal{D}_{d,T} = \left\{ \xi^T_{ijk} = \frac{i v_1 + j v_2 + k v_3}{d} \ :\ i + j + k = d \right\}, \tag{3}$$

and every $p \in \mathcal{B}_d(T)$ has a unique B-form $p = \sum_{i+j+k=d} c^T_{ijk} B^d_{ijk}$ with the B-coefficients c^T_{ijk} attached to the domain points ξ^T_{ijk}. For a triangulation $\triangle$ we write $\mathcal{D}_{d,\triangle} = \bigcup_T \mathcal{D}_{d,T}$, and the space of piecewise elements with no continuity imposed is $S_d^{-1}(\triangle)$, of dimension $N = \binom{d+2}{2}|\triangle|$; its elements are in bijection with the vectors $c = (c^T_{ijk})$ of B-coefficients, which we call the broken coefficient vector. The spline spaces are

$$S^r_d(\triangle) = \{s \in C^r : s|_T \in \mathcal{B}_d(T) \text{ for all } T \in \triangle\} \subset S_d^{-1}(\triangle). \tag{4}$$

Membership in $S^r_d(\triangle)$ is a set of linear conditions on c, one family per interior edge. Let $e = \langle v_2, v_3\rangle$ be shared by $T = \langle v_1, v_2, v_3\rangle$ and $\widetilde{T} = \langle v_4, v_3, v_2\rangle$, and write c^T_{ijk} and $c^{\widetilde{T}}_{ijk}$ for the two sets of B-coefficients. Then s is C^r across e if and only if

$$c^{\widetilde{T}}_{\rho jk} \;=\; \sum_{\nu+\mu+\kappa=\rho} c^T_{\nu,\, j+\mu,\, k+\kappa}\, B^\rho_{\nu\mu\kappa}(v_4), \qquad \rho = 0, \dots, r, \quad j+k = d-\rho, \tag{5}$$

where $B^\rho_{\nu\mu\kappa}(v_4)$ is the Bernstein basis polynomial of degree ρ associated with T, evaluated at the fourth vertex v_4 [32, Thm. 2.28] in the plane and [32, Thm. 13.30] on the sphere; the hyperboloid is the same computation with Q_κ in place of the Euclidean form, since the proof lifts the two pieces to homogeneous polynomials on the cones over T and $\widetilde{T}$ and matches their derivatives across the plane through the origin and e. In the plane the barycentric coordinates of v_4 sum to one and (5) is the classical condition; on the sphere and the hyperboloid they do not, and (5) holds unchanged. The condition for $\rho = 0$ is the equality of the $d+1$ B-coefficients on e, that is continuity, and each further ρ ties the ρ-th row of coefficients of $\widetilde{T}$ parallel to e to the first $\rho+1$ rows of T. The only geometric input is the position of v_4 in the barycentric coordinates of T.

Stacking (5) over all interior edges and all $\rho \le r$ gives the smoothness matrix $J = J^{(r)}_d(\triangle) \in \mathbb{R}^{m\times N}$, so that

$$S^r_d(\triangle) = \{\, s \in S^{-1}_d(\triangle) : Jc = 0 \,\}, \qquad \dim S^r_d(\triangle) = N - \operatorname{rank} J. \tag{6}$$

The rows of J are not independent, since the conditions around a vertex are related, and no attempt is made to select an independent subset; the rank is computed. A matrix Z with range $Z = \ker J$ and full column rank is built once by sparse rank-revealing elimination in a locality order [24, 25], so that $c = Z\hat{c}$ parametrizes the spline space by unconstrained coefficients $\hat{c}$. In the language of [32, Ch. 5] the columns of Z extend the coefficients on a minimal determining set of $S^r_d(\triangle)$ to all of $\mathcal{D}_{d,\triangle}$, and the locality order is what keeps that extension local, hence Z sparse with a level-independent stencil (Section 7.1).

3.3 Gradients and the model Laplacian

Assembly needs the gradient of a Bernstein polynomial and, for the fourth-order remarks, the model Laplacian; both are algebraic in the coefficients. Let P be the homogeneous degree-d polynomial on $\mathbb{R}^3$ whose restriction to X_κ is the spline piece, written in the Bernstein form (2) with $b = V^{-1}x$. Differentiating in x,

$$\nabla_x P(x) = V^{-\top} \begin{bmatrix} \partial P/\partial b_1 \\ \partial P/\partial b_2 \\ \partial P/\partial b_3 \end{bmatrix}, \tag{7}$$

by the chain rule with $b = V^{-1}x$, and each $\partial P/\partial b_i$ is again a Bernstein form, of degree $d-1$, whose coefficients are $d\, c_{\beta+e_i}$ for $|\beta| = d-1$, the factor d coming from $\partial B^d_\alpha/\partial b_i = d\, B^{d-1}_{\alpha-e_i}$; so the map from the coefficients of P to those of its three partial derivatives is a sparse matrix with one nonzero entry per row. The vector (7) is the Euclidean coordinate gradient of the homogeneous extension of P to $\mathbb{R}^3$. The tangential gradient on the model is the ambient gradient with its normal part removed,

$$\nabla_X P(x) = \big(Q_\kappa - \kappa\, x\, x^\top\big)\nabla_x P(x), \qquad x \in X_\kappa,\ \kappa = \pm 1, \tag{8}$$

obtained by raising the index with $Q_\kappa^{-1} = Q_\kappa$, which turns the coordinate gradient into the gradient for the form $\langle\cdot,\cdot\rangle_\kappa$, and then removing the normal component along x, the projector onto the tangent plane $\{\xi : \langle x, \xi\rangle_\kappa = 0\}$ being $\xi \mapsto \xi - \kappa\, x\langle x, \xi\rangle_\kappa$. For $\kappa = 1$ this is $(I - xx^\top)\nabla_x P$, the usual spherical

gradient; for $\kappa = -1$ it is $(Q_{-1} + xx^\top)\nabla_x P$, which is tangent to the hyperboloid and whose inner product with itself under $\langle\cdot,\cdot\rangle_{-1}$ is positive. For $\kappa = 0$ the plane $x_0 = 1$ is flat and the planar gradient is the last two components of (7). The stiffness integrand of (24) is $\langle \nabla_X B^d_\alpha, \nabla_X B^d_\beta \rangle_X$ with the inner product of Q_κ, evaluated at the quadrature points. The model Laplace–Beltrami operator acts on the Bernstein form by

$$\Delta_{X_\kappa} P = \operatorname{tr}\big(W \operatorname{Hess}_b P\big) - \kappa\, d(d+1)\, P, \qquad W = V^{-1} Q_\kappa V^{-\top}, \tag{9}$$

with no metric evaluation and no Christoffel symbols; the sign of the zeroth-order term is fixed by the degree-one eigenfunctions of the model, $\Delta_{\mathbb{S}^2} x_3 = -2x_3$ on the sphere and $\Delta_{\mathbb{H}^2} x_0 = +2x_0$ on the hyperboloid, and the term is absent in the plane, where (9) is the classical Bernstein Laplacian.

3.4 Integration: closed forms on the plane, quadrature on the hyperboloid

On the plane every integral in the assembly has a closed form. Products of Bernstein polynomials are Bernstein polynomials, $B^d_\alpha B^p_\gamma = \binom{\alpha+\gamma}{\alpha}\binom{d+p}{d}^{-1} B^{d+p}_{\alpha+\gamma}$ with multi-index binomials, and $\int_T B^m_\mu \, dA = |T| / \binom{m+2}{2}$, so

$$\int_T B^d_\alpha B^d_\beta \, dA = \frac{|T|}{\binom{2d}{d}\binom{2d+2}{2}} \binom{\alpha+\beta}{\alpha}, \tag{10}$$

and the stiffness integrand, a product of two Bernstein forms of degree $d-1$ by (7), is of the same kind. A weight enters in closed form once it is a Bernstein form: replacing w by its C^0 Bernstein interpolant $\widehat{w}$ of degree p on the same triangulation makes $\int_T \widehat{w} B^d_\alpha B^d_\beta$ a sum of terms (10) with $2d+p$ in place of $2d$, and the load vector with $\widehat{wf}$ likewise. The sums over the weight's coefficients factorize over the three barycentric components, and the Bernstein moment recursion of Ainsworth, Andriamaro and Davydov [1] assembles the weighted matrices in $O(d^4)$ operations per triangle, the number of entries; see also Kirby [30]. The flat experiments of Section 8 were run with the Gauss rule below and reproduced with this closed-form assembly, and the difference between the two is the interpolation error of the weight, reported in Section 8.3 as the δ of Proposition 6.2(iii).

On the hyperboloid no closed form is available. In the Klein chart, the central projection $x \mapsto y = (x_1/x_0, x_2/x_0)$ of the upper sheet onto the unit disk, geodesics are straight chords and geodesic triangles are Euclidean triangles, but a Bernstein form is $(1-|y|^2)^{-d/2}$ times a polynomial and the area density is $(1-|y|^2)^{-3/2}$, so the mass integrand is a polynomial times $(1-|y|^2)^{-d-3/2}$ and the stiffness integrand carries the same algebraic factor; these are not elementary integrals. They are computed by a conical Gauss product rule, a tensor rule on the square mapped to the reference triangle by the Duffy transformation [18], with q points per direction. The integrand is analytic on the polygon and its nearest singularity lies on the unit circle, so the rule converges geometrically, with relative error $O(\rho^{-2q})$ for a $\rho > 1$ fixed by the distance from the Klein-chart image of F to the circle; the density is steep near the boundary of the polygon, so the coarse triangles need a higher-order rule than their size suggests, the convergence of the rule is reported in Table 3, and $q = d + 4$ points per direction are used unless stated otherwise. Two details of the rule matter for this integrand. The collapsed vertex of the Duffy map is placed at the corner of the triangle farthest from the origin, where the density is largest, since the map clusters its points there. And Gauss–Legendre is kept in the collapsed direction: the Jacobian $(1-u)$ of the map is a Jacobi weight and could be absorbed by a Gauss–Jacobi rule [28], which gains a degree for a smooth integrand, but on the Bolza octagon the Legendre rule gives an area defect two to three times smaller at every q from 8 to 32, because its points lie closer to the corner where the integrand grows; the criterion for this integrand is the distance of the nodes to the singularity, and the Jacobi

rule moves them away from it. The remark after Proposition 6.2 says how the quadrature error propagates to the computed spectrum.

4 Topology: fundamental polygons and their identifications

This section contains everything that depends on the surface's topology and nothing that depends on its geometry. Its output is the smoothness matrix J of the quotient, and its check is the Gauss–Bonnet theorem, applied in the model metric where no conformal factor has yet entered.

4.1 Flat quotients: torus, Klein bottle, Möbius strip

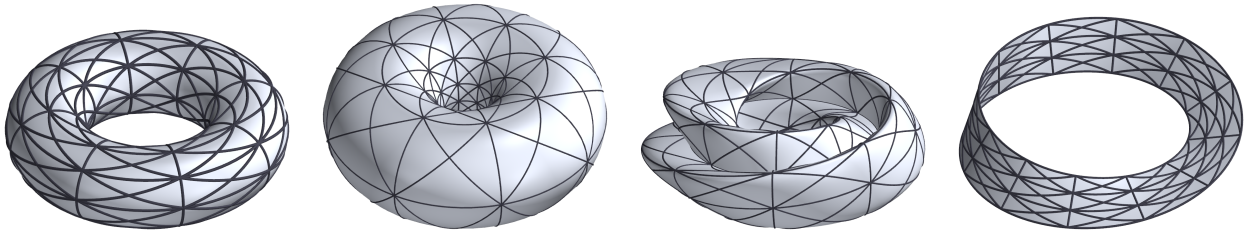
Let $F = [0, a] \times [0, b]$ in the plane. A side pairing of a polygon F is an involution σ on its sides together with, for each side s, an isometry γ_s of the model carrying s onto $\sigma(s)$ and F onto the copy of F on the far side of $\sigma(s)$; identifying x with $\gamma_s x$ for every s produces the quotient surface, and the group Γ generated by the γ_s acts on the model with F as a fundamental domain. Written as maps of the plane, the identifications are

$$\begin{array}{lll} \text{torus} & \gamma_1(x_1, x_2) = (x_1 + a,\ x_2), & \gamma_2(x_1, x_2) = (x_1,\ x_2 + b), \\ \text{Klein bottle} & \gamma_1(x_1, x_2) = (x_1 + a,\ x_2), & \gamma_2(x_1, x_2) = (a - x_1,\ x_2 + b), \\ \text{Möbius strip} & \gamma_1(x_1, x_2) = (x_1 + a,\ b - x_2), & x_2 = 0,\ b \text{ free}, \end{array} \tag{11}$$

so that γ_1 pairs the left side with the right and γ_2 the bottom with the top. The maps γ_2 of the Klein bottle and γ_1 of the Möbius strip are glide reflections, compositions of a translation with a reflection in a line parallel to it; they reverse orientation, which is what makes those two surfaces nonorientable. The group Γ is generated by the pairings, and F is a fundamental domain for its action on Ω, its Γ-images covering Ω, the whole plane for the torus and the Klein bottle and the strip $\mathbb{R} \times [0, b]$ for the Möbius band, and meeting only along their boundaries. The triangulations used below are criss-cross meshes: the square is divided into $n \times n$ congruent cells and each cell into four triangles by its two diagonals, a choice symmetric under the reflections appearing in (11) and therefore compatible with the pairings in the sense of Section 4.3. The Euler characteristic is zero in all three cases, the torus and Klein bottle being closed and the Möbius strip having one boundary circle. Figure 1 shows the torus and the Klein bottle as spline surfaces in $\mathbb{R}^3$, and Figure 2 the identified square with its periodic triangulation; the mesh is a structured triangulation of F whose boundary vertices correspond under (11), which for the reflecting pairings means a mesh symmetric under $x_1 \mapsto a - x_1$ or $x_2 \mapsto b - x_2$.

4.2 Hyperbolic quotients: side-paired polygons

Let $\Gamma < \mathrm{Isom}(\mathbb{H}^2)$ be torsion free, discrete and cocompact [29, Ch. 6]. If $\Gamma \subset SO^+(2, 1)$ the quotient is orientable of genus $g \geq 2$ and Γ is a Fuchsian group; otherwise the quotient is nonorientable. A geodesic fundamental polygon F with n sides and a side pairing σ, an involution on the sides together with an isometry $\gamma_s \in \Gamma$ carrying side s onto side $\sigma(s)$, presents the quotient when the hypotheses of the Poincaré polygon theorem hold [36, Ch. 13]: the pairings match endpoints, and around every cycle of identified corners the interior angles sum to 2π and the product of the pairings along the cycle is the identity. The regular octagon with opposite sides paired, all angles $\pi/4$, presents the Bolza surface of genus two [8]; a regular 14-gon with Klein's pairing presents the Klein quartic of genus three [31]; a hexagon with the pairing word $aabbcc$ presents the nonorientable surface N_3 of Euler characteristic -1 (Figure 4). The orientable ones embed smoothly in $\mathbb{R}^3$, a genus-two surface



torus of revolution, $R = 2$, $r = 0.8$, modulus $\tau = 0.436\,i$ | Clifford torus, $R = r\sqrt{2}$, $\tau = i$ | Klein bottle, figure-eight immersion | Möbius band, one boundary circle

Figure 1: The flat quotients with the criss-cross mesh of the square carried through an embedding or immersion of $\mathbb{R}^3$, for display. The computation takes place on the square F with the identifications (11), and the surface enters only through the conformal factor of Section 5; the two tori are embedded and their conformal factors are given by (19), the Klein bottle and the Möbius band admit no embedding in $\mathbb{R}^3$ and are drawn as an immersion and a half-twisted band.

as a pretzel, and the nonorientable N_3 immerses but does not embed; none embeds isometrically: a closed surface in $\mathbb{R}^3$ has a point of positive Gauss curvature, at which a circumscribed sphere touches it, so by Hilbert's theorem no closed hyperbolic surface admits a C^2 isometric embedding [17, Ch. 5]. The figures therefore show the fundamental domain on the hyperboloid rather than a surface in space, and Figure 3 shows what the surface finite element method meshes instead; the computation never uses an embedding.

The pairing pattern, not only the polygon, fixes the surface: the 14-gon with opposite sides paired is a different genus-three surface from the Klein quartic [27].

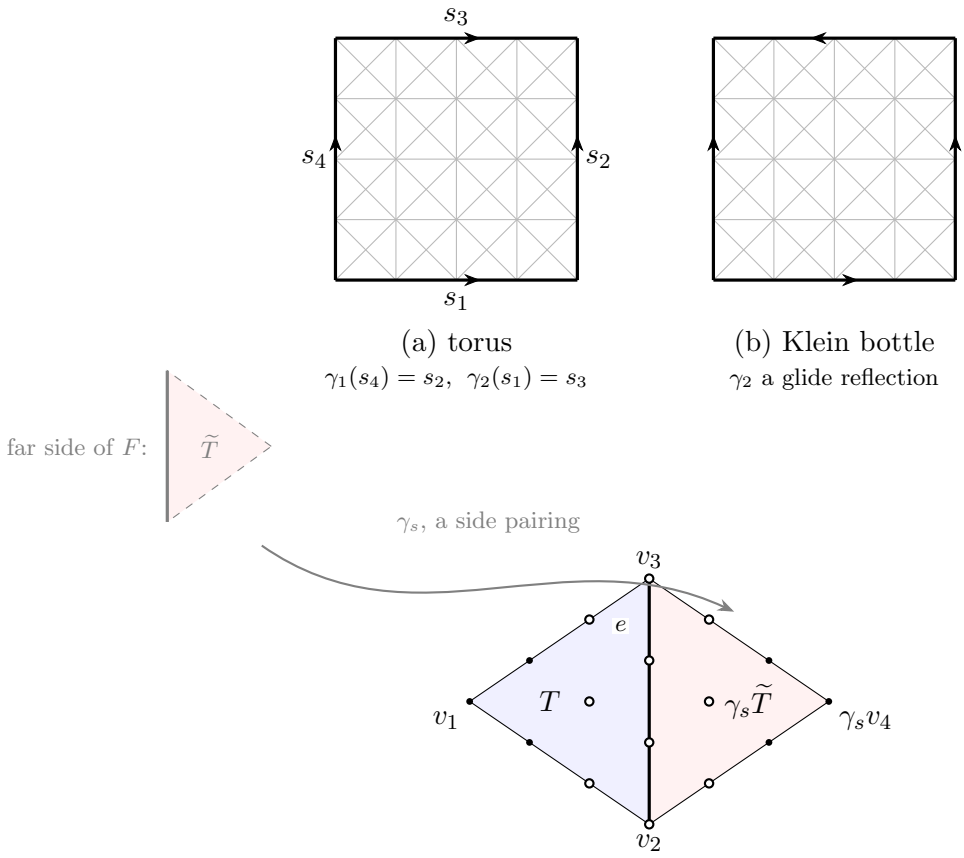
The corner cycle is computed by a walk: from a corner, cross the side to its right by the pairing, turn to the next side, and repeat until the starting corner returns; the composite of the pairings along the walk must be the identity of $O^+(2,1)$. Two conventions matter, the order of composition, which symmetry hides on the octagon and Klein's pairing exposes by ten orders of magnitude, and a direction flag once orientation-reversing pairings are admitted [27]; both are caught by the checks of Section 4.6.

4.3 Meshes that respect the identifications

A triangulation $\triangle$ of F is pairing compatible if paired sides carry matching subdivisions under their pairings. The Γ-translates of $\triangle$ then triangulate X, and the domain points on paired sides correspond under the pairings. On the plane the structured triangulations of a rectangle are compatible with the translations, and the symmetric ones with the glide reflections. On the hyperboloid, refinement by joining geodesic midpoints preserves compatibility, because isometries preserve geodesics and midpoints, so the domain and its identifications are represented without approximation at every level (Figure 6); no polygonal approximation of the surface is ever introduced, in either model.

4.4 Other fundamental domains

The input to the construction below is a triangulation of the quotient together with an isometry across each interior edge, and a side-paired polygon is one way to supply it, with the isometries being deck transformations of the polygon's sides. It is not the only way. A closed hyperbolic surface of genus $g \geq 2$ decomposes along $3g - 3$ disjoint simple closed geodesics into $2g - 2$ pairs of pants, and each pair of pants into two isometric right-angled hexagons [13, Ch. 3]; the surface



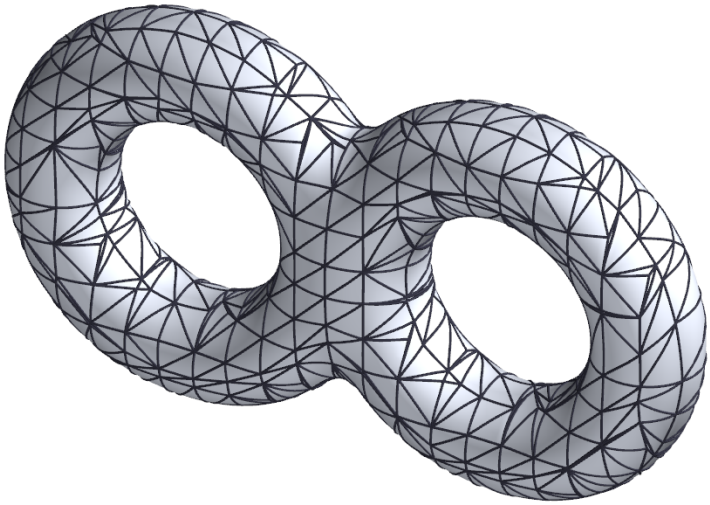
(a) torus
$\gamma_1(s_4) = s_2,\ \gamma_2(s_1) = s_3$

(b) Klein bottle
γ_2 a glide reflection

(c) the conditions (5) across a paired edge, $d = 3$, $r = 1$

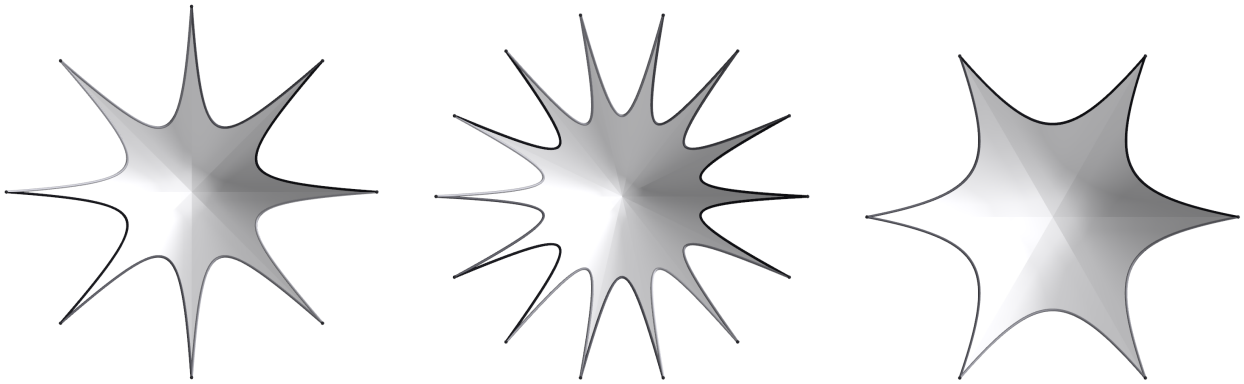
$$c^{\gamma_s\widetilde{T}}_{\rho jk} = \sum_{\nu+\mu+\kappa=\rho} c^{T}_{\nu,\, j+\mu,\, k+\kappa}\, B^{\rho}_{\nu\mu\kappa}(\gamma_s v_4), \qquad \rho = 0, \dots, r, \quad j + k = d - \rho$$

Figure 2: Topology as rows of the smoothness matrix. (a) and (b): the square F with its criss-cross triangulation $\triangle$ and the side pairings (11), whose only difference is the direction of one arrow; reversing it turns the torus into the Klein bottle and the group into one containing a glide reflection. (c): the conditions (5) across a paired edge, drawn with the domain points $\mathcal{D}_{3,T}$ and the shared edge $e = \langle v_2, v_3\rangle$. The neighboring triangle $\widetilde{T}$ sits at the far side of F and is transported by the pairing γ_s before the conditions are written; by Proposition 4.1 its B-coefficients are carried unchanged, so the paired rows have the same form as interior ones and the isometry enters only through the position of $\gamma_s v_4$. Open circles mark the coefficients tied by the C^1 conditions, the two rows nearest the edge in T and the first two rows in $\gamma_s\widetilde{T}$.



a pretzel in $\mathbb{R}^3$, genus two, with a surface-conforming mesh of
585 vertices and 1184 curved triangles

Figure 3: A closed genus-two surface embedded in $\mathbb{R}^3$ with the metric induced from space, meshed. This is the setting of the surface finite element method [19, 20, 9], in which the surface is approximated by the mesh and a geometric consistency error enters the analysis. The hyperbolic quotients of this section carry curvature -1 and admit no C^2 isometric embedding, so a picture of this kind exists for the pretzel and not for the Bolza surface.

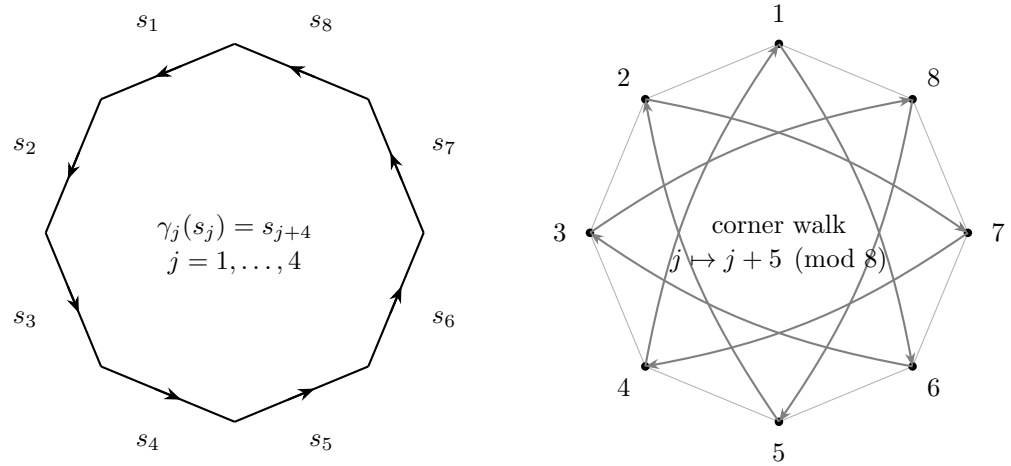


Bolza surface, genus 2: octagon, opposite sides paired

Klein quartic, genus 3: 14-gon, Klein's pairing

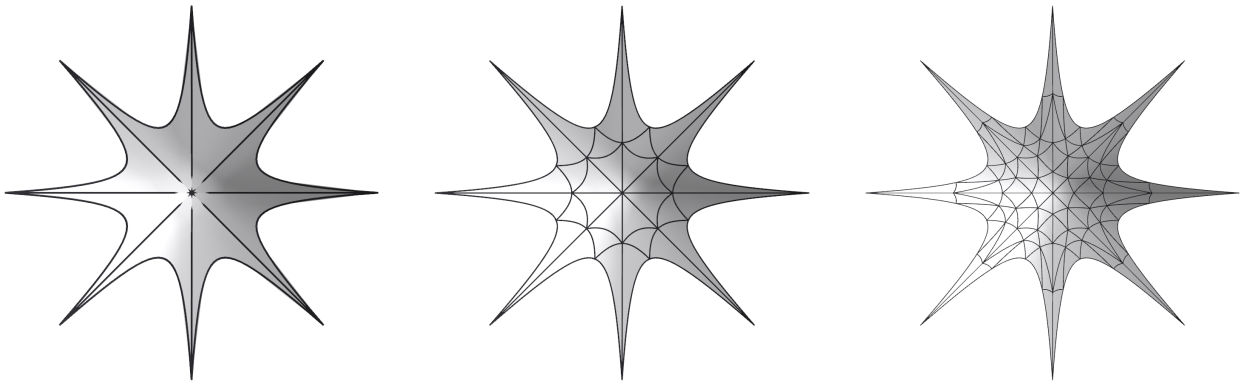
N_3, nonorientable, $\chi = -1$: hexagon, word *aabbcc*

Figure 4: Fundamental polygons on the hyperboloid, viewed along the x_0 axis in orthographic projection so that the geodesic sides bow toward the apex. The pairing pattern, not only the polygon, fixes the surface.



(a) the Bolza octagon and its side pairing (b) one vertex cycle of length eight

Figure 5: The genus-two computation. (a) The regular hyperbolic octagon with interior angles $\pi/4$ and its side pairing: each side is carried onto the opposite one by a hyperbolic translation of length $2\,\mathrm{arccosh}(1+\sqrt{2})$, and the four translations generate Γ. (b) The corner walk of Section 4.2. Since $\gcd(5,8) = 1$ the walk $j \mapsto j+5$ visits all eight corners in one cycle, their angles sum to $8 \cdot \pi/4 = 2\pi$ as the Poincaré polygon theorem requires [36, Ch. 13], and the product of the pairings along the cycle is the identity of $O^+(2,1)$, which is checked numerically to 3.4×10^{-11}. The quotient is a closed orientable two-manifold of genus two, that is a sphere with two handles, with $\chi = -2$ and hyperbolic area $-2\pi\chi = 4\pi$.



level 0: 8 triangles level 1: 32 triangles level 2: 128 triangles

Figure 6: Pairing-compatible geodesic refinement of the Bolza octagon at three levels. Every drawn curve is a geodesic of the surface and the boundary subdivisions correspond under the side pairings.

is then $4g-4$ hexagons glued along their sides by isometries, and the gluing is described by the $6g-6$ Fenchel–Nielsen coordinates, the lengths of the $3g-3$ geodesics and the twists along them. A right-angled hexagon is determined by three alternate side lengths, half the boundary lengths of its pair of pants, and Figure 7 shows one hexagon, a pair of pants and four hexagons around a common vertex, a fundamental domain of a genus-two surface. Every interior angle is $\pi/2$ at every genus, each piece sits near the origin of its own chart, and the conformal class is an explicit input; these are the three properties the regular $4g$-gon lacks at large genus (Section 8.6). Only the mesh layer changes: the rows of Section 4.5 are written across each glued side against the neighbor transported by the gluing isometry, in the same way as across a paired side of a polygon.

That the polygon and the group are inessential is tested. Table 1 presents the Bolza surface from its combinatorial triangulation and hyperbolic edge lengths alone: each triangle is placed in $\mathbb{H}^2$ from its three edge lengths, the isometry across each interior edge is computed from the two placements, and assembly proceeds as in Section 6. The transitions carry shared vertices across to 6×10^{-14}, the area is 4π to 10^{-14}, and μ_1 agrees with the octagon computation of Table 10 to all digits at every degree and level. The same layer accepts the edge lengths of a pants decomposition, which is how a surface given by Fenchel–Nielsen coordinates or by a discrete uniformization enters; the hexagon presentation of Figure 7 is stated here as the input format for that companion computation and is not run through the eigenvalue pipeline in this paper.

Table 1: The Bolza surface presented as an intrinsic triangulation: faces, shared edges and hyperbolic edge lengths only, with no fundamental polygon and no group. Each triangle is placed in $\mathbb{H}^2$ separately and the transition isometry across each edge is computed from the two placements; the residual column is the largest mismatch of a shared vertex under its transition. The eigenvalues coincide with the octagon computation of Table 10 to the digits shown.

level	d	$\dim S^0_d$	transition residual	area defect	$\mu_{0,h}$	μ_1 (rel. error)
1	4	254	8.9×10^{-14}	3.6×10^{-14}	7.8×10^{-15}	3.852134233 (3.5×10^{-3})
1	6	574	8.9×10^{-14}	3.4×10^{-14}	1.6×10^{-14}	3.839143822 (6.7×10^{-5})
2	4	1022	6.1×10^{-14}	3.6×10^{-15}	9.5×10^{-15}	3.838936644 (1.3×10^{-5})
2	6	2302	6.1×10^{-14}	7.1×10^{-15}	2.1×10^{-14}	3.838887323 (1.7×10^{-8})

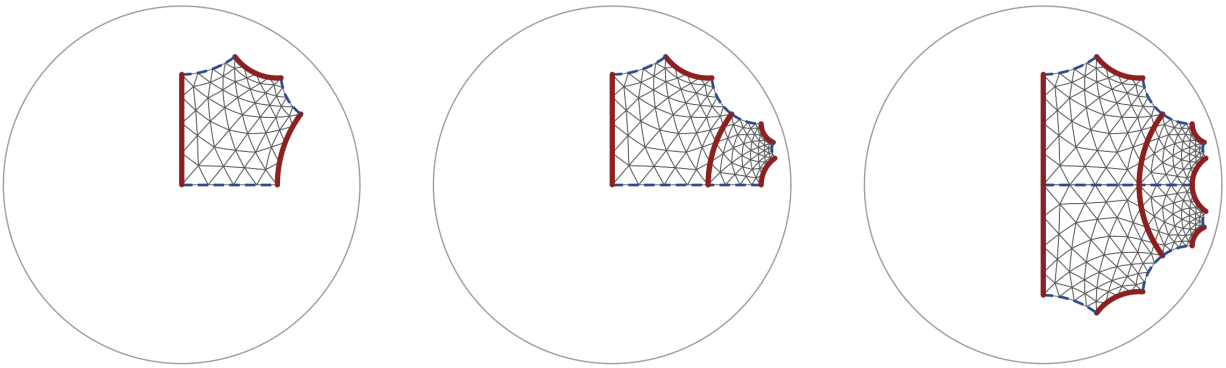
4.5 The identifications as rows of the smoothness matrix

The single fact that makes the quotient computation identical in form to the computation on a domain is the following.

Proposition 4.1 (Pairings act trivially on coefficients)**.** *Let $T\subset X_\kappa$ be a geodesic triangle with vertex matrix V and γ an isometry of X_κ, acting linearly on $\mathbb{R}^3$. Then the barycentric coordinates of γx with respect to γT are those of x with respect to T, and if $s|_T=\sum_\alpha c_\alpha B^d_\alpha(\,\cdot\,;T)$ then $(s\circ\gamma^{-1})|_{\gamma T}=\sum_\alpha c_\alpha B^d_\alpha(\,\cdot\,;\gamma T)$.*

Proof. $Vb=x$ implies $(\gamma V)b=\gamma x$, and γV is the vertex matrix of γT. □

A side pairing therefore carries a triangle's coefficient vector unchanged to the image triangle. Across a paired edge the C^r conditions (5) are imposed in the same way as across an interior edge, with the neighboring triangle first transported by the pairing, so that the fourth vertex entering (5) is $\gamma_s v_4$ and the B-coefficients of $\widetilde{T}$ are used unchanged, by Proposition 4.1. This is the whole topological content of the discretization.



a right-angled hexagon, alternate sides $a = b = c = 1.2$, fan mesh refined twice

a pair of pants: the hexagon and its reflection across a seam

four hexagons around a vertex: a genus-two fundamental domain

Figure 7: The pants decomposition in the Poincaré disk. Seams, along which the two hexagons of a pair of pants are glued, are drawn as thick solid lines (dark red); the halves of the boundary geodesics, along which pairs of pants are glued to each other with a twist, are drawn as thinner dashed lines (blue), so the two kinds of side remain distinct in black and white. All interior angles are $\pi/2$; the hexagon closes under its six right-angle turns to 6×10^{-13}.

Theorem 4.2 (Quotient splines)**.** *Coefficient arrays on the triangles of a pairing-compatible triangulation $\triangle$ of F that satisfy (5) across interior edges and across each paired edge after transport by its pairing are in one-to-one correspondence with Γ-invariant C^r splines on the lifted triangulation of Ω, the Γ-invariant region of X covered by the images of F, hence with C^r splines on the quotient Ω/Γ; for a closed surface $\Omega = X$, and for the Möbius band Ω is the strip $\mathbb{R} \times [0, b]$. The statement holds whether or not Γ contains orientation-reversing isometries.*

Proof. Transport every coefficient vector to all Γ-images of its triangle by Proposition 4.1; different group words reaching the same triangle give the same geometric triangle and the same coefficients because the pairings present a genuine quotient. Interior edges of every copy satisfy (5), and paired edges do so by hypothesis, so the lifted function is C^r and Γ-invariant. Conversely the restriction of a quotient spline to F satisfies both families. Linearity of the isometries is all that is used, so orientation plays no role. □

In matrix terms the smoothness matrix of the quotient is the interior matrix with the paired-edge rows appended, its kernel is the quotient spline space $S^r_d(X/\Gamma, \triangle)$, and the null-space matrix Z is built from it in the same way as on a domain. Two features of the rows are worth recording. The pairing never enters a matrix entry: on the Bolza octagon at degree four the barycentric coordinates of the transported fourth vertex are $(-1, 1, 1)$, the same numbers as for a flat fold, although the pairing isometry has Frobenius norm 21.3, because the neighbor is transported before those coordinates are computed [27]. And the rows around an identified vertex are dependent, two per vertex cycle on a closed surface, which is where the topology shows in the rank.

Proposition 4.3 (Dimension of the quotient spaces)**.** *Let $\triangle$ be a pairing-compatible triangulation of F with V, E, T vertices, edges and triangles counted on the quotient, so that $V - E + T = \chi$. Then*

$$\dim S^0_d = V + (d-1)E + \binom{d-1}{2} T, \qquad \operatorname{rank} J^{(0)} = N - \dim S^0_d, \tag{12}$$

and, for $d \geq 5$,

$$\dim S^1_d = \binom{d-1}{2} T + 3\chi + \sigma, \tag{13}$$

where σ is the number of singular vertices, those at which the edges lie on two lines, each of which adds one dimension.

Proof. For $r = 0$ the space consists of the continuous piecewise elements, and a continuous piecewise polynomial is determined by its values at the domain points of the quotient triangulation, each counted once; the count of distinct domain points is V vertices, $d-1$ interior points on each edge and $\binom{d-1}{2}$ interior points on each triangle, which is (12). The identification enters only through the fact that V and E are counted on the quotient, that is, cycles of identified vertices and pairs of identified edges count once.

For $r = 1$ and $d \geq 5$ the generic dimension of a planar C^1 space is known [32, Ch. 9], and the local nature of the smoothness conditions makes the count on a quotient the same computation as on a simply connected patch, with the Euler relation of the quotient in place of the planar one. Writing the total number of coefficients as $\binom{d+2}{2}T$, the number of C^1 rows as $(2d+1)E$ and the dependencies as three per vertex cycle, plus one more at each singular vertex, one at which all edges lie on two straight lines, and using $V - E + T = \chi$ and $3T = 2E - E_\partial$ gives (13) after simplification, with the boundary term of the corollary below. The identity is a statement about the generic rank of the constraint system, so we treat it as a prediction and compare it with the computed rank in Table 2 rather than assuming it. □

Corollary 4.4 (Surfaces with boundary)**.** *If the quotient has a boundary consisting of E_∂ edges, the dimension formula* (13) *acquires the term $(d-1)E_\partial$, the boundary edges carrying no smoothness condition.*

On the criss-cross mesh of the square with $n \times n$ cells, every cell center is a vertex at which the four edges lie on two straight lines, that is, a singular vertex in the sense of Morgan and Scott [34], [32, Ch. 9], so $\sigma = n^2$; this is the reason the criss-cross mesh is used here, since it makes the C^1 space larger and the comparison with the identity sharper. The Möbius band has $E_\partial = 2n$ boundary edges. Table 2 compares the computed dimension $N - \operatorname{rank} J$ with the prediction on all three flat quotients and on the two hyperbolic ones. The identity holds at every configuration with $n \geq 4$; at $n = 2$ the mesh is too coarse for the generic count, a single cell then having two of its sides identified with each other, and the computed dimension is larger, which is the expected behavior of dimension formulas on degenerate configurations and is reported rather than hidden.

4.6 The Gauss–Bonnet checks

The Gauss–Bonnet theorem, $\int_M K\, dA = 2\pi\chi(M)$, touches every layer of the construction and is applied in the model metric, before any conformal factor enters, at three levels.

Combinatorial. On the quotient triangulation $V - E + T = \chi$, with V counting vertex cycles of the polygon rather than its corners. This tests the identification bookkeeping alone.

Angles. For a geodesic triangulation of F in the hyperbolic metric, $\sum_T (\pi - \alpha_T - \beta_T - \gamma_T) = \operatorname{Area}(F) = -2\pi\chi$, and at every vertex cycle the angles sum to 2π. The first tests the side pairings and the second the group parameters: with wrong parameters the pairings still close combinatorially but a vertex cycle has the wrong angle sum. On the flat quotients both sums are identities of the plane, and the check reduces to the closing of the vertex cycles.

Reference curvature by quadrature. $\int_F K_X\, dA_X = 2\pi\chi$ with the assembly quadrature and the area density of the model, which tests the quadrature against the closed-form angle result.

Table 2: Dimension of the quotient spline spaces: the computed value $N - \operatorname{rank} J$ from the rank-revealing elimination against the prediction of Proposition 4.3 and Corollary 4.4. Flat quotients on the criss-cross mesh of the square with $n \times n$ cells, so $T = 4n^2$, $\sigma = n^2$ and, for the Möbius band, $E_\partial = 2n$; hyperbolic quotients on the geodesic mesh at level L, where $\sigma = 0$. The C^0 prediction is (12) with the counts taken on the quotient.

surface	mesh	d	r	χ	computed	predicted	prediction
torus	$n = 4$	2	0	0	128	128	$32 + 96 + 0$
	$n = 8$	4	0	0	2048	2048	$128 + 3 \cdot 384 + 3 \cdot 256$
	$n = 4$	6	1	0	656	656	$10 \cdot 64 + 0 + 16$
	$n = 8$	6	1	0	2624	2624	$10 \cdot 256 + 0 + 64$
Klein bottle	$n = 4$	2	0	0	128	128	as the torus
	$n = 4$	6	1	0	656	656	$10 \cdot 64 + 0 + 16$
	$n = 8$	6	1	0	2624	2624	$10 \cdot 256 + 0 + 64$
Möbius band	$n = 4$	4	0	0	528	528	domain points
	$n = 4$	6	1	0	696	696	$10 \cdot 64 + 16 + 5 \cdot 8$
	$n = 8$	6	1	0	2704	2704	$10 \cdot 256 + 64 + 5 \cdot 16$
Bolza	$L = 1$	4	0	-2	574	574	$14 + 3 \cdot 48 + 3 \cdot 32$
	$L = 1$	6	1	-2	314	314	$10 \cdot 32 - 6$
	$L = 2$	6	1	-2	1274	1274	$10 \cdot 128 - 6$
N_3	$L = 1$	6	1	-1	237	237	$10 \cdot 24 - 3$

Table 3 reports the three rungs for the five quotients of this paper. On the flat quotients $K_X = 0$, so the curvature integral is zero as $2\pi\chi = 0$ requires, and the second and third rungs reduce to the closing of the vertex cycles: all four corners of the square form one cycle of angle sum 2π on the torus and the Klein bottle, and two boundary vertices of angle π on the Möbius strip. On the Bolza octagon the eight corners form one cycle with angle sum 2π, the area by angle defect is 4π to 2×10^{-14}, and the Klein-chart quadrature of the hyperbolic area density converges to it, 3.6×10^{-2}, 4.8×10^{-4}, 1.4×10^{-7} and 4.5×10^{-11} at 8, 16, 32 and 48 points per direction, the steep density $(1 - |y|^2)^{-3/2}$ near the polygon boundary being the reason a high-order rule is needed on the coarse triangles [27]. The two remaining rungs, the flux identity for a spline conformal factor and the curvature of an embedded spline surface, need a metric and are stated in Section 5.

Table 3: Metric-free Gauss–Bonnet checks in the model metric g_X. V, E, F are counted on the quotient; the angle column gives the angle sum of each corner cycle of the polygon; the curvature column is $\int_F K_X \, dA_X$, which Gauss–Bonnet requires to equal $2\pi\chi$ and which vanishes on the flat quotients because $K_X = 0$ there; the last column is the area of F in g_X, equal to $-2\pi\chi$ on the hyperbolic quotients since $K_X = -1$, and to $ab = 1$ for the unit square. For the hyperbolic quotients the area is given by angle defect and by Klein-chart quadrature at 48 points per direction, the two defects in parentheses.

surface	polygon	V	E	T	χ	corner cycles	$\int_F K_X \, dA_X$	area of F
torus	square, 4×4	32	96	64	0	one, 2π	0	1
Klein bottle	square, 4×4	32	96	64	0	one, 2π	0	1
Möbius band	square, 4×4	36	100	64	0	two boundary, π each	0	1
Bolza	octagon, $L = 1$	14	48	32	-2	one, 2π	-4π	4π (10^{-14}, 10^{-11})
N_3	hexagon *aabbcc*	2	3	1	-1	two, 2π each	-2π	2π (10^{-15})

4.7 A first result: the closed-form spectra of the genus-one quotients

The flat quotients have closed-form spectra, so with the model metric alone the discrete eigenvalue problem $a(u_h, v_h) = \mu_h\, b(u_h, v_h)$ on $S^r_d(X/\Gamma, \triangle)$ tests the topological layer with no geometry in it. On the unit-square torus the eigenvalues are $4\pi^2(m^2+n^2)$; on the Klein bottle of (11) with $a = b = 1$ they are $4\pi^2 m^2 + \pi^2 n^2$ with n even for the cosine modes in x_1 and n odd for the sine modes; on the Möbius band with free sides they are $\pi^2(m^2+n^2)$ with $m+n$ even. At $d = 4$ and at $(d, r) = (6, 1)$ on the criss-cross mesh the computed eigenvalues reproduce these to all printed digits, with their multiplicities: $4\pi^2$ four times and $8\pi^2$ four times on the torus, $4\pi^2$ three times and $5\pi^2$ twice on the Klein bottle, $2\pi^2$ twice and $4\pi^2$ three times on the Möbius band; at $d = 2$ the same multiplicities appear with errors of 2×10^{-4}, the approximation error of eigenfunctions that are not polynomials. The multiplicity three of $4\pi^2$ on the Klein bottle, against four on the torus, is the glide reflection acting on the spectrum, and it is the first place the paired rows across an orientation-reversing pairing are seen to be right.

5 Geometry: the conformal factor as data

This section is short by design. The geometry of the surface enters the computation as one positive function on the polygon, and the only properties of it that the discretization and the analysis use are the two identities of Lemma 5.1 and the bounds $w_{\min}$, $w_{\max}$, $\|w\|_{C^1}$ of Corollary 7.2.

Everything in Section 4 was done in the model metric g_X. The surface M of (1) carries $\bar g = e^{2\lambda} g_X$ with λ a Γ-invariant smooth function, that is, a function on the polygon F whose values and derivatives match across paired sides; any element of the spline space of Section 4.5 has this property, which is how prescribed factors are represented below. We write $w = e^{2\lambda}$.

5.1 Mass and energy of the conformal metric

The area element, gradient and Laplace–Beltrami operator of $\bar g$ are

$$dA_M = e^{2\lambda}\, dA_X, \qquad \nabla_M u = e^{-2\lambda} \nabla_X u, \qquad \Delta_M u = e^{-2\lambda} \Delta_X u, \tag{14}$$

the last the conformal transformation law in dimension two [23, Ch. 2]. Write $\langle \xi, \eta \rangle_X$ for the inner product of two tangent vectors in the model metric: the Euclidean dot product on the plane, and the restriction of $\langle \xi, \eta \rangle_{-1}$ to the tangent plane on the hyperboloid, where it is positive definite; $\langle \xi, \eta \rangle_M$ denotes the inner product of $\bar g = e^{2\lambda} g_X$. On the polygon define

$$a(u, v) := \int_F \langle \nabla_X u, \nabla_X v \rangle_X \, dA_X, \qquad b_w(u, v) := \int_F w\, u\, v \, dA_X, \tag{15}$$

for Γ-invariant u, v; both are integrals over the quotient since F is a fundamental domain. The gradient $\nabla_X u$ and the inner product $\langle \cdot, \cdot \rangle_X$ both belong to the model, so a is the Dirichlet form of X and contains no λ.

Lemma 5.1 (Mass and energy)**.** *For all $u, v \in W^{1,2}(M)$,*

$$\int_M u\, v \, dA_M = b_w(u, v), \qquad \int_M \langle \nabla_M u, \nabla_M v \rangle_M \, dA_M = a(u, v), \tag{16}$$

and $E^{-1}\|u\|_{L^2(X/\Gamma)} \le \|u\|_{L^2(M)} \le E\|u\|_{L^2(X/\Gamma)}$ with $E = e^{\max_F |\lambda|}$; in particular $W^{1,2}(M) = W^{1,2}(X/\Gamma)$ with equivalent norms.

Proof. The first identity is the first of (14). For the second, which is where the dimension enters, since the two powers of $e^{2\lambda}$ cancel only when the domain is two-dimensional, $\langle \nabla_M u, \nabla_M v\rangle_M$ equals $e^{-2\lambda}\langle \nabla_X u, \nabla_X v\rangle_X$ pointwise by the second, and the factor cancels against dA_M. The norm comparison is $E^{-2} \le w \le E^2$. □

The energy is therefore assembled in the model metric once, and the surface enters the computation only through w at the quadrature points. On the flat quotients the model energy is the flat Dirichlet integral over the square. On a hyperbolic quotient the same cancellation is applied a second time: in the Klein chart $y = (x_1/x_0, x_2/x_0)$ of the hyperboloid the polygon becomes a Euclidean polygon with straight sides, the hyperbolic metric is $\rho(y)^2$ times a flat metric only in the Poincaré chart, but the Dirichlet energy is chart independent, so

$$a(u,v) = \int_{F_P} \nabla u \cdot \nabla v \, dy, \qquad b_w(u,v) = \int_{F_P} w \left(\frac{2}{1-|y|^2}\right)^2 u\, v \, dy \tag{17}$$

in the Poincaré chart F_P, and the hyperbolic area density joins w in the weight. In the implementation the integrals are evaluated on the hyperboloid through the Klein chart, where geodesic triangles are Euclidean triangles, with the tangential gradient of Section 3.3; the statement (17) is what makes the hyperbolic stiffness matrix a flat matrix in disguise and is the reason the same solver serves all genera.

5.2 Normalizations

On a hyperbolic quotient the group Γ fixes the conformal structure, so there is no Möbius freedom in λ; the only normalization is the scale, and we do not impose one, the hyperbolic area $-2\pi\chi$ of the model serving as the reference. On the torus the conformal structure is fixed by the modulus of the defining parallelogram, the complex number τ whose ratio of sides it records, two flat tori being conformally equivalent if and only if their moduli agree; here the parallelogram is the unit square and $\tau = i$, and λ is determined up to an additive constant which we fix by $\int_F w\, dA_X = \mathrm{Area}(F)$ in the experiments that compare spectra. On the Klein bottle and the Möbius strip the same holds with the glide.

5.3 Instances

A prescribed factor. Any Γ-invariant smooth λ defines an instance; the experiments use $\lambda = 0.3\cos 2\pi x_1 + 0.2\sin 2\pi x_2$ on the torus, $\lambda = 0.3\cos 2\pi x_1 \cos 2\pi x_2$ on the Klein bottle, which is invariant under the glide $(x_1, x_2) \mapsto (1 - x_1, x_2 + 1)$, and a spline λ on the Bolza surface.

An embedded torus of revolution. Let $\Sigma \subset \mathbb{R}^3$ be the torus with radii $R > r > 0$, parametrized by $(\theta, \phi) \mapsto ((R + r\cos\theta)\cos\phi,\ (R + r\cos\theta)\sin\phi,\ r\sin\theta)$, whose induced metric is $r^2 d\theta^2 + (R + r\cos\theta)^2 d\phi^2$. Introducing the meridian variable

$$u(\theta) = \int_0^\theta \frac{r\, dt}{R + r\cos t} = \frac{2r}{\sqrt{R^2 - r^2}} \arctan\Big(\sqrt{\tfrac{R-r}{R+r}}\, \tan\tfrac{\theta}{2}\Big), \qquad 0 \le \theta < \pi, \tag{18}$$

extended to $[0, 2\pi)$ by $u(\theta) = U - u(2\pi - \theta)$, the continuous branch of the integral, makes the metric $(R + r\cos\theta)^2(du^2 + d\phi^2)$, so Σ is the flat torus $[0, U] \times [0, 2\pi]$ with $U = u(2\pi) = 2\pi r/\sqrt{R^2 - r^2}$, carrying the conformal factor

$$w(u, \phi) = \big(R + r\cos\theta(u)\big)^2, \tag{19}$$

where $\theta(u)$ inverts (18) in closed form. The conformal class of Σ is that of the rectangle of modulus $\tau = iU/2\pi = i\, r/\sqrt{R^2 - r^2}$, and the Clifford torus is the case $\tau = i$, that is $R = r\sqrt{2}$. The tori of

Figure 1 are drawn from this data: the conformal factor (19) is what enters the computation, and the embedding is used only to display the surface.

A hyperbolic surface with a prescribed factor. The Bolza surface with λ a spline of degree six and smoothness one on the quotient, fixed at the coarsest level and carried to the finer meshes by midpoint subdivision, so that the metric does not change with the mesh.

A surface given as a mesh. The pair (Γ, λ) must then be computed by discrete uniformization; that instance is the subject of a companion paper.

5.4 The remaining Gauss–Bonnet rungs

The curvature of $\bar{g}$ is $\bar{K} = e^{-2\lambda}(K_X - \Delta_X \lambda)$ with $K_X \in \{0, -1\}$, so

$$\int_M \bar{K}\, dA_M = K_X \operatorname{Area}(F) - \int_F \Delta_X \lambda \, dA_X, \tag{20}$$

so for a closed quotient Gauss–Bonnet reduces to $\int_F \Delta_X \lambda \, dA_X = 0$, and for a quotient satisfying Assumption 6.1 the remaining flux is the boundary geodesic-curvature term. The integral is not automatically zero for an arbitrary function on the polygon, and this is what makes it a test.

Proposition 5.2 (Flux identity)**.** *Let $\lambda \in S^1_d(X/\Gamma, \triangle)$. Then*

$$\int_F \Delta_X \lambda \, dA_X = \sum_{T \in \triangle} \int_{\partial T} \frac{\partial \lambda}{\partial n}\, ds = \int_{\partial M} \frac{\partial \lambda}{\partial n}\, ds_X, \tag{21}$$

which vanishes when X/Γ is closed, so that (20) reduces to $\int_M \bar{K}\, dA_M = 2\pi\chi$ for every spline conformal factor of smoothness at least one. Under Assumption 6.1 the boundary of M has geodesic curvature $\bar{k}_g = e^{-\lambda}\, \partial\lambda/\partial n$ in $\bar{g}$, so that $\int_{\partial M} \bar{k}_g\, ds_M = \int_{\partial M} \partial\lambda/\partial n\, ds_X$ and (20) becomes the Gauss–Bonnet identity with boundary, $\int_M \bar{K}\, dA_M + \int_{\partial M} \bar{k}_g\, ds_M = 2\pi\chi$.

Proof. The first equality is the divergence theorem on each triangle, λ being a polynomial there. In the sum, each interior edge appears twice with opposite outward normals, and the two normal derivatives agree because λ is C^1 across it, so the contributions cancel. Each boundary edge of F appears once, but its partner appears once as well, and the pairing is an isometry carrying the one edge to the other and reversing the outward normal; since λ is Γ-invariant its normal derivative is carried along, so those contributions cancel in pairs. The free edges, if any, are what remains, and their union is ∂M. □

On a closed quotient the identity is a test of the paired rows alone. On a quotient with boundary the free sides of F contribute the flux $\int_{\partial M} \partial\lambda/\partial n\, ds_X$, which is the geodesic curvature term of the boundary in $\bar{g}$: the sides are geodesics of the model but not of $\bar{g}$ unless $\partial\lambda/\partial n = 0$ on them, the conformal change of geodesic curvature being $\bar{k}_g = e^{-\lambda}(k_g + \partial\lambda/\partial n)$ with $k_g = 0$ here. The check on the Möbius band is therefore that the computed curvature integral and the computed boundary flux sum to $2\pi\chi = 0$.

The identity holds in the continuum and up to quadrature error in the computation, so its measured value tests the paired C^1 rows themselves: it fails by a computable flux if a pairing is misoriented, if a row across an identified edge is missing, or if the corner walk composes in the wrong order, and each of those errors leaves the C^0 merging and the dimension count intact.

6 The discrete problems

6.1 Variational forms

Let $f \in L^2(M)$ with $\int_M f\, dA_M = 0$. By Lemma 5.1 the weak form of the Poisson equation $-\Delta_M u = f$ with $\int_M u\, dA_M = 0$ is the problem on the polygon in the model metric: find $u \in W^{1,2}(X/\Gamma)$ with $b_w(u,1) = 0$ such that

$$a(u,v) = b_w(f,v) \qquad \text{for all } v \in W^{1,2}(X/\Gamma), \tag{22}$$

and the eigenvalue problem $-\Delta_M u = \mu u$ is the pencil

$$a(u,v) = \mu\, b_w(u,v) \qquad \text{for all } v \in W^{1,2}(X/\Gamma). \tag{23}$$

Both are well posed. The form b_w is an inner product equivalent to the L^2 inner product of the model by Lemma 5.1, and a is coercive on the closed subspace $\{v : b_w(v,1) = 0\}$ of the closed quotient, since there $a(v,v) \geq \mu_1(M)\, b_w(v,v)$ with $\mu_1(M) > 0$ the first nonzero eigenvalue; the Lax–Milgram lemma [12, Ch. 2] gives a unique solution of (22), which then holds for all v because both sides vanish at $v = 1$. The solution operator of (23) is compact and self-adjoint in b_w, so the spectrum is real and discrete, $0 = \mu_0 < \mu_1 \leq \mu_2 \leq \cdots \to \infty$ [14, Ch. I]. The quotients with boundary are admitted under the following standing assumption, which the Möbius band and the cylinder satisfy.

Assumption 6.1 (Quotients with boundary)**.** The free sides of F form, on the quotient, finitely many closed curves that are geodesics of the model metric g_X without corners, and the Laplace–Beltrami operator is taken with the natural boundary condition $\partial u/\partial n = 0$ on ∂M, where n is the outward normal of $\bar{g}$.

Under Assumption 6.1 the weak forms (22) and (23) are unchanged, since the natural condition imposes nothing on the trial and test spaces, the constraint $b_w(u,1) = 0$ is kept for the Poisson problem, and the spectrum is the Neumann spectrum of M; the normal derivative is the same for $\bar{g}$ and g_X up to the factor $e^{-\lambda}$, so the condition is $\partial u/\partial n = 0$ in the model as well. On the Möbius band the single boundary circle is the image of the two free sides $x_2 = 0, b$, joined by the glide into one closed straight line of the plane, so the assumption holds; the cylinder has two such circles. The statements above hold with the same proofs, elliptic regularity for the Neumann problem on a compact manifold with smooth boundary taking the place of regularity on a closed manifold [40, Ch. 5].

6.2 The discrete problems in broken coefficients

Let $S_h = S^r_d(X/\Gamma, \triangle)$ be the space of Theorem 4.2, $N = \binom{d+2}{2}|\triangle|$ the number of broken coefficients, J the smoothness matrix with its interior and paired rows, and Z with range $Z = \ker J$. On each triangle T, with the quadrature points $x_q \in T$ and weights ω_q of Section 3.4 and the Bernstein basis (2), the element matrices and vectors are

$$\begin{aligned} (K_T)_{\alpha\beta} &= \sum_q \omega_q \,\langle \nabla_X B^d_\alpha(x_q), \nabla_X B^d_\beta(x_q)\rangle_X, & (M_{w,T})_{\alpha\beta} &= \sum_q \omega_q\, w(x_q)\, B^d_\alpha(x_q) B^d_\beta(x_q), \\ (F_T)_\alpha &= \sum_q \omega_q\, w(x_q)\, f(x_q)\, B^d_\alpha(x_q), & (c_1)_\alpha &= \text{coefficients of } 1, \end{aligned} \tag{24}$$

and $K = \operatorname{diag}_T K_T$, $M_w = \operatorname{diag}_T M_{w,T}$ are block diagonal, one block per triangle, with F and c_1 the concatenations. Three features of (24) carry the paper. The stiffness matrix K contains no w,

so it is assembled once per $(\triangle, d)$ and reused for every metric on that quotient. The topology is not in K or M_w either, only in J; the broken matrices are the same for the torus and the Klein bottle on the same mesh. And c_1 exists if and only if the constants lie in the space: on the flat quotients at every degree, on the hyperbolic ones for even d only, by Section 3.

The discrete Poisson problem is: find $c \in \ker J$ with $c_1^\top M_w c = 0$ and $y^\top K c = y^\top F$ for all $y \in \ker J$. With $c = Z\hat{c}$ and a scalar multiplier ν this is the bordered symmetric system

$$\begin{bmatrix} Z^\top K Z & g \\ g^\top & 0 \end{bmatrix} \begin{bmatrix} \hat{c} \\ \nu \end{bmatrix} = \begin{bmatrix} Z^\top F \\ 0 \end{bmatrix}, \qquad g = Z^\top M_w c_1, \tag{25}$$

whose matrix is nonsingular because $Z^\top K Z$ is positive semidefinite with kernel spanned by the reduced coefficients of the constant and $g^\top \hat{c}_1 = \int_F w \, dA_X > 0$. The discrete eigenvalue problem is the reduced pencil

$$(Z^\top K Z)\, \hat{c} = \mu_h \, (Z^\top M_w Z)\, \hat{c}, \tag{26}$$

with $Z^\top M_w Z$ positive definite, solved by shift-invert Lanczos [33] at a small negative shift, which maps the low end of the spectrum including $\mu_0 = 0$ to dominant separated eigenvalues. The multiplier ν of (25) is the discrete compatibility defect and is reported as a solver check; the residual of (25) must be read after projection by $Z^\top$, since the broken residual $Kc - F$ is a Lagrange multiplier in range $J^\top$ and is $O(1)$ even for the discrete solution itself.

6.3 Three structural facts

The following hold for every $\triangle$, d and r, and are used as checks throughout Section 8.

Proposition 6.2. *Let $S_h \subset W^{1,2}(X/\Gamma)$ and let $\mu_{i,h}$ be the eigenvalues of (26).*

(i) One-sidedness. *$\mu_{i,h} \geq \mu_i$ for $i = 0, \ldots, \dim S_h - 1$.*

(ii) Reproduced kernel. *If the constants lie in S_h then $(1, 0)$ is a discrete eigenpair, so $\mu_{0,h} = 0$.*

(iii) Uniform weight perturbation. *If a positive function $\widehat{w}$ is used in place of w and $\delta = \|\widehat{w}/w - 1\|_{L^\infty} < 1$, that is $(1-\delta)\, w \leq \widehat{w} \leq (1+\delta)\, w$ pointwise, so that $(1-\delta)\, b_w \leq b_{\widehat{w}} \leq (1+\delta)\, b_w$, then the eigenvalues $\widehat{\mu}_{i,h}$ of the pencil with mass form $b_{\widehat{w}}$ satisfy $\mu_{i,h}/(1+\delta) \leq \widehat{\mu}_{i,h} \leq \mu_{i,h}/(1-\delta)$ for every i.*

Proof. (i) is the Courant–Fischer principle [14, Ch. I] applied over subspaces of S_h; (ii) is $a(1, v_h) = 0 = \mu\, b_w(1, v_h)$; (iii) follows from $(1-\delta)b_w \leq b_{\widehat{w}} \leq (1+\delta)b_w$ as quadratic forms and the monotonicity of the Courant–Fischer minimum in the mass form. □

Part (iii) is a statement about replacing one weight by another, for instance an approximate conformal factor by the one it approximates, and its content is that the resulting eigenvalue error is relative and uniform along the spectrum. Quadrature is a different perturbation and is treated separately. On the flat quotients there are two ways to assemble, both in Section 3.4. With the closed-form assembly the computed pencil is the Galerkin pencil for the interpolated weight $\widehat{w}$, integrated without error, so (i) holds for it and (iii) with $\delta = \|\widehat{w}/w - 1\|_{L^\infty} = O(h^{p+1})$ brackets it against the pencil for w; nothing is left to observation. With the Gauss rule only the mass form is approximated, the stiffness integrand being polynomial; the assembled mass form is a discrete inner product b_h on S_h, and if $(1 - \delta_h)\, b_w \leq b_h \leq (1 + \delta_h)\, b_w$ the argument of (iii) gives the same bracket with $\delta_h = O(h^{2q})$ for a positive rule of order q and smooth w [38, Ch. 4], [5]. On the hyperbolic

quotients both forms are approximated by the rule, with relative errors δ_K and δ_M of order ρ^{-2q} by Section 3.4, and the same argument gives for the computed pencil

$$\mu_{i,h}\,\frac{1-\delta_K}{1+\delta_M} \;\le\; \widehat{\mu}_{i,h} \;\le\; \mu_{i,h}\,\frac{1+\delta_K}{1-\delta_M}, \tag{27}$$

so the computed eigenvalues lie within a relative distance $O(\rho^{-2q})$ of the Galerkin ones, uniformly along the spectrum; the computed pencil is not itself the conforming Rayleigh–Ritz pencil of (i), so one-sidedness of the computed values is observed in Section 8 and holds up to (27). The quadrature order is chosen so that the area is reproduced to roundoff, which places δ_M below the eigenvalue errors reported. Part (ii) fails at odd degree on a hyperbolic quotient, where the constants are not in the space, and then the whole spectrum is polluted; this is the reason for the parity restriction and is measured in Section 8. On the Bolza octagon at level 1 the computed zero mode is 6.7×10^{-16} at $d = 4$ and 2.0×10^{-14} at $d = 6$, and 1.9×10^{-3} at $d = 3$ and 3.3×10^{-5} at $d = 5$, where the constants are approximated rather than represented.

6.4 Algorithm

Algorithm 1 Poisson equation and eigenvalues on a quotient surface

Require: fundamental polygon F with its side pairings; conformal factor w; right-hand side f; degree d and smoothness r; refinement level

Ensure: broken coefficient vector c of u_h, or the eigenpairs of (26), with the outputs of the checks: Euler characteristic and corner-cycle angle sums, $N - \operatorname{rank} J$ against (12) or (13), area defect, zero mode

1: **Topology.** Check the pairings: matched endpoints, corner cycles closing with angle sum 2π and identity product, Euler characteristic (Section 4.6).

2: **Mesh.** Triangulate F compatibly with the pairings and refine by midpoints; the identifications are preserved (Section 4.3).

3: **Constraints, once per** $(\triangle, d, r)$**.** Assemble J from the interior and paired edges by (5) with the neighbor transported by its pairing; obtain Z by rank-revealing elimination; compare $N - \operatorname{rank} J$ with (12) or (13).

4: **Stiffness, once per** $(\triangle, d)$**.** Assemble K from (24) and form $Z^{\top} K Z$; neither depends on w.

5: **Weighted data.** Evaluate w and f at the quadrature points and assemble $M_w c_1$, F, and for the eigenproblem M_w.

6: **Solve.** (25) by a sparse direct solver, or (26) by shift-invert Lanczos; record ν, the projected residual, and, on the flat quotients where the stiffness matrix is integrated without error, the one-sidedness of Proposition 6.2; on the hyperbolic quotients record the position of the computed eigenvalues relative to the reference as an observation.

7 Error estimates

The rates observed in Section 8 follow from a local approximation property of the spline space on the quotient together with the standard conforming arguments. We state the hypotheses, because two of them are specific to quotients: the approximated function is Γ-invariant, and the weight is bounded above and below on the polygon rather than on a chart.

Theorem 7.1 (Approximation on the quotient)**.** *Let $M = \Omega/\Gamma$ be closed, or have a boundary satisfying Assumption 6.1. Let $\triangle$ be a pairing-compatible triangulation of F with mesh size $h = \max_{T\in\triangle} \operatorname{diam} T$ and smallest angle bounded below, let $d \geq 3r+2$, and let $v \in W^{d+1,2}(X/\Gamma)$. Then there is $s \in S^r_d(X/\Gamma, \triangle)$ with*

$$\|v - s\|_{L^2(X/\Gamma)} + h\,|v-s|_{1,2,X/\Gamma} \;\leq\; C\,h^{\,d+1}\,|v|_{d+1,2,X/\Gamma}, \tag{28}$$

with C depending on d, r and the shape regularity, and for $X = \mathbb{H}^2$ also on the diameter of F in the Klein chart.

Proof. Both cases are proved by constructing a quasi-interpolant on the quotient from a stable local determining set, and the only point that is special to a quotient is that such a set exists there.

By Theorem 4.2 the space $S^r_d(X/\Gamma, \triangle)$ is the space of Γ-invariant elements of $S^r_d(\widetilde{\triangle})$ on the lifted triangulation of X, and the lift of a paired edge of F is an interior edge of $\widetilde{\triangle}$. A stable local determining set for $S^r_d(\widetilde{\triangle})$ exists for $d \geq 3r+2$ [32, Thm. 5.24 and Ch. 8], on a triangulation with or without boundary, it can be chosen Γ-invariantly because Γ acts on $\widetilde{\triangle}$ by simplicial isomorphisms preserving the construction, and its image in F is then a determining set for the quotient space, with the same locality and the same stability constant. Let Q_h be the associated quasi-interpolant: it reproduces $\mathcal{B}_d$ on each star, is bounded on L^∞ of the star with a constant depending only on d, r and the smallest angle, and is local, $Q_h v|_T$ depending only on v on $\operatorname{star}^\ell(T)$ for a fixed $\ell = \ell(d,r)$.

For $X = \mathbb{E}^2$ the pieces of S^r_d are ordinary bivariate polynomials, and the Bramble–Hilbert lemma [12, Ch. 4] on each star, combined with the locality and boundedness of Q_h and summation over $\triangle$, gives (28) in the form $\|v - Q_h v\|_{L^2} + h|v - Q_h v|_{1,2} \leq Ch^{d+1}|v|_{d+1,2}$, which is the classical estimate [32, Ch. 10] with the stars now taken on the quotient.

For $X = \mathbb{H}^2$ the pieces are restrictions of homogeneous polynomials, not polynomials in a chart. Pass to the Klein chart $y = (x_1/x_0, x_2/x_0)$, in which a geodesic triangle is a Euclidean triangle and

$$B^d_\alpha(x)\big|_y = \big(1 - |y|^2\big)^{-d/2}\, p_\alpha(y), \qquad p_\alpha \in \mathcal{P}_d,$$

so that the chart image of S^r_d is $\omega_d \mathcal{P}_d$ pieces with the single fixed weight $\omega_d(y) = (1-|y|^2)^{-d/2}$, smooth and bounded above and below on the compact set $\overline{F_K}$, the chart image of the polygon, together with all its derivatives. Apply the planar argument to v/ω_d and multiply back: the norms of v and v/ω_d are equivalent on $\overline{F_K}$ with constants depending on d and on $\max_{\overline{F_K}}|y| < 1$, that is on the diameter of F in the chart, and the hyperbolic and Euclidean volume elements and gradients are likewise comparable there. This gives (28) with the stated constant. The argument is the one of [27], where the dependence of the constant on $\max|y|$ is examined; it degrades as the polygon approaches the ideal boundary, which for the fixed surfaces of this paper is a fixed constant. □

Corollary 7.2 (Rates)**.** *Let X/Γ be closed or satisfy Assumption 6.1, let $w \in C^\infty$ with $0 < w_{\min} \leq w \leq w_{\max}$ on F, let the hypotheses of Theorem 7.1 hold, and let u solve (22) with $u \in W^{d+1,2}(X/\Gamma)$. Then*

$$|u - u_h|_{1,2,M} \leq Ch^{\,d}\,|u|_{d+1,2,X/\Gamma}, \qquad \|u - u_h\|_{L^2(M)} \leq Ch^{\,d+1}\,|u|_{d+1,2,X/\Gamma}, \tag{29}$$

with C depending in addition on $w_{\min}$, $w_{\max}$ and $\|w\|_{C^1}$. If μ is an eigenvalue of (23) whose eigenspace lies in $W^{d+1,2}(X/\Gamma)$, the discrete eigenvalues converging to μ satisfy $0 \leq \mu_h - \mu \leq Ch^{\,2d}$.

Proof. The energy estimate is Céa's lemma on the w-mean-zero subspace, where a is coercive by Section 6.1, together with (28) and the identity $a =$ the energy of M from Lemma 5.1. For the L^2 estimate let $e = u - u_h$ made w-mean-free and let z solve $-\Delta_X z = we$ with $b_w(z,1) = 0$; elliptic regularity on the closed manifold X/Γ, or for the Neumann problem on the compact manifold

with smooth boundary of Assumption 6.1, [40, Ch. 5] gives $\|z\|_{W^{2,2}} \le C w_{\max} \|e\|_{L^2}$, Galerkin orthogonality gives $b_w(e,e) = a(z - z_h, e)$ for every $z_h \in S_h$, and (28) with d replaced by 1 bounds the best approximation of z in energy by $Ch\|z\|_{W^{2,2}}$; dividing by $\|e\|_{L^2_w} \ge w_{\min}^{1/2} \|e\|_{L^2}$ and using the energy estimate gives the L^2 rate, which Lemma 5.1 transfers to M. The eigenvalue statement is the Babuška–Osborn estimate for conforming approximation of a symmetric pencil with compact solution operator [4, Sec. 7]: the eigenvalue error is bounded by the square of the best energy approximation of the eigenspace, which (28) bounds by h^d. One-sidedness is Proposition 6.2(i). □

Two remarks distinguish this from the surface finite element setting. There is no geometric consistency term: the polygon and its identifications are represented without approximation at every refinement level (Section 4.3), so the only variational crime is quadrature: on the flat quotients of the mass form alone, the stiffness integrand being polynomial, and on the hyperbolic quotients of both forms, as discussed after Proposition 6.2. And for a fixed quotient X/Γ and mesh family, the conformal factor enters the constants of (29) only through $w_{\min}$, $w_{\max}$ and $\|w\|_{C^1}$; the dependence on the quotient itself, through the shape regularity, the Klein-chart diameter of F in the hyperbolic case and the coercivity constant $\mu_1(M)$, is that of the underlying conforming method and is not uniform over all quotients, as the genus ladder of Section 8.6 shows. The rates themselves are the same on every quotient, which is what Section 8 observes on the torus, the Klein bottle and the Möbius band.

7.1 Conditioning and cost

The estimates above concern the solution of (25) and (26), and the constrained form raises the question of whether the chosen locally supported null-space basis preserves the refinement scaling expected of a second-order problem. Table 4 examines this for the torus on the criss-cross mesh, reporting the spectral condition number of the reduced operator $Z^\top(K + M_w)Z$, which is the relevant quantity for the source problem, and of the reduced mass matrix $Z^\top M_w Z$, which governs the eigensolve. The first grows like h^{-2}, the rate of a second-order problem in any conforming space: the measured factors per halving of h are 3.99 and 4.00 at $d = 2$ and 3.98 and 4.00 at $d = 6$, $r = 1$. The second is independent of h, as it must be for a mass matrix in a locally supported basis, and grows with the degree, by a factor near 20 per step of two in d, which is the conditioning of the Bernstein basis itself and not an effect of the constraints.

The cost of the constraint layer is also independent of the refinement: Z has 3.00 and 4.90 nonzeros per column at $(d, r) = (2, 0)$ and $(6, 1)$ and $Z^\top K Z$ has 11.5 and 55.2 per row, identical at every level, so the reduction generates no level-dependent fill, and the assembly and elimination are performed once per $(\triangle, d, r)$ and reused for every conformal factor on that quotient.

8 Numerical experiments

The surfaces, by genus: in genus one, with the flat plane as model, the torus, the Klein bottle and the Möbius band (Sections 4.7 to 8.3); in genus two and above, with the hyperbolic plane as model, the Bolza surface and the regular-polygon surfaces of the genus ladder (Sections 8.5 and 8.6). Every subsection heading names its genus. The order follows the two channels: the topology alone, through closed-form spectra; the geometry, through the Poisson problem and eigenvalues on conformal quotients; then the hyperbolic case, where both are present and the reference is external. All genus-one computations use the criss-cross mesh of the unit square with $n \times n$ cells, conical Gauss quadrature with $d + 4$ points per direction, the bordered null-space system of Section 6, and the

Table 4: Conditioning and sparsity on the torus, criss-cross mesh with $n \times n$ cells and the flat metric. The condition number of $Z^\top(K + M_w)Z$ grows like h^{-2}; that of the reduced mass matrix is independent of h. The nonzero counts per column of Z and per row of $Z^\top KZ$ do not change with the level.

(d, r)	n	$\dim S_h$	$\mathrm{nnz}(Z)/\mathrm{col}$	$\mathrm{nnz}(Z^\top KZ)/\mathrm{row}$	$\kappa(Z^\top(K+M_w)Z)$	$\kappa(Z^\top M_w Z)$
$(2, 0)$	4	128	3.00	11.5	6.43×10^2	3.14×10^1
	8	512	3.00	11.5	2.56×10^3	3.14×10^1
	16	2048	3.00	11.5	1.03×10^4	3.14×10^1
$(6, 1)$	4	656	4.90	55.2	7.16×10^3	9.51×10^3
	8	2624	4.90	55.2	2.85×10^4	9.58×10^3
	16	10496	4.90	55.2	1.14×10^5	9.60×10^3

errors of Lemma 5.1, the $L^2(M)$ error being the w-weighted L^2 error on the square and the $H^1(M)$ error the flat Dirichlet error.

8.1 Genus one: the Poisson equation on the conformal torus and Klein bottle

Test 1 of the genus-zero paper is repeated on the torus and the Klein bottle: the solution u is prescribed on the quotient, not a polynomial, and $f = -\Delta_M u = -e^{-2\lambda}\Delta_{\mathbb{E}^2} u$ is computed symbolically, so that the product wf in the load vector is $-\Delta_{\mathbb{E}^2} u$ and contains no factor; the metric then enters only the constraint vector and the norm of the error. Table 5 gives the conformal torus and Table 6 the conformal Klein bottle. The $H^1(M)$ rates are 2.0, 4.0 and 6.1 for $d = 2, 4, 6$ and the $L^2(M)$ rates 3.0, 5.0 and 7.2 on both quotients; the energy errors of the flat and the conformal torus agree to the digits shown, as Lemma 5.1 predicts for a load vector without the factor, the multiplier ν is below 10^{-13} in every row, and the Klein bottle rows are the first in which a glide reflection carries a nontrivial metric through the C^1 rows.

Table 5: Poisson Test 1 on the torus $[0, 1]^2/\mathbb{Z}^2$ with the conformal metric $e^{2\lambda} g_{\mathbb{E}^2}$, $\lambda = 0.3 \cos 2\pi x_1 + 0.2 \sin 2\pi x_2$; prescribed solution $u = \sin 2\pi x_1 \cos 4\pi x_2 + \frac{1}{2} \cos 2\pi(x_1 + x_2)$, criss-cross mesh with $n \times n$ cells. Errors in $L^2(M)$ and the $H^1(M)$ seminorm with rates over the last refinement. The same computation with the flat metric gives the same energy errors to the digits shown, since the load vector of Test 1 contains $-\Delta_{\mathbb{E}^2} u$ and no factor.

(d, r)	n	$\dim S_h$	$L^2(M)$ (rate)	$H^1(M)$ (rate)
$(2, 0)$	4	128	4.32×10^{-2}	1.66×10^{0}
	8	512	5.97×10^{-3} (2.85)	4.13×10^{-1} (2.01)
	16	2048	8.01×10^{-4} (2.90)	1.07×10^{-1} (1.95)
	32	8192	1.02×10^{-4} (2.97)	2.69×10^{-2} (1.99)
$(4, 0)$	4	512	1.04×10^{-3}	6.64×10^{-2}
	8	2048	3.18×10^{-5} (5.03)	3.97×10^{-3} (4.06)
	16	8192	1.03×10^{-6} (4.95)	2.54×10^{-4} (3.97)
	32	32768	3.24×10^{-8} (4.99)	1.60×10^{-5} (3.99)
$(6, 1)$	4	656	3.30×10^{-5}	2.22×10^{-3}
	8	2624	1.59×10^{-7} (7.69)	2.38×10^{-5} (6.54)
	16	10496	1.09×10^{-9} (7.19)	3.43×10^{-7} (6.12)

Table 6: Poisson Test 1 on the Klein bottle with the conformal metric $e^{2\lambda}g_{\mathbb{E}^2}$, $\lambda = 0.3\cos 2\pi x_1 \cos 2\pi x_2$, which is invariant under the glide; prescribed solution $u = \sin 2\pi x_1 \sin \pi x_2 + \cos 2\pi x_1 \cos 2\pi x_2$, a smooth function on the Klein bottle.

(d,r)	n	h^{-1}	$\dim S_h$	conformal Klein bottle	
				$L^2(M)$ (rate)	$H^1(M)$ (rate)
$(2,0)$	4	4	128	1.11×10^{-2}	4.08×10^{-1}
	8	8	512	1.51×10^{-3} (2.88)	1.07×10^{-1} (1.94)
	16	16	2048	1.93×10^{-4} (2.96)	2.70×10^{-2} (1.98)
	32	32	8192	2.43×10^{-5} (2.99)	6.76×10^{-3} (2.00)
$(4,0)$	4	4	512	7.15×10^{-5}	4.84×10^{-3}
	8	8	2048	2.33×10^{-6} (4.94)	3.11×10^{-4} (3.96)
	16	16	8192	7.35×10^{-8} (4.98)	1.96×10^{-5} (3.99)
	32	32	32768	2.30×10^{-9} (5.00)	1.23×10^{-6} (4.00)
$(6,1)$	4	4	656	3.57×10^{-7}	3.20×10^{-5}
	8	8	2624	2.44×10^{-9} (7.20)	4.61×10^{-7} (6.12)
	16	16	10496	1.83×10^{-11} (7.06)	7.03×10^{-9} (6.04)

8.2 Genus one with boundary: the eigenvalue problem on the conformal Möbius band

The Möbius band is the quotient with a boundary, covered by Assumption 6.1 since its boundary is one closed straight line of the plane, and computed as the Neumann problem of Section 6.1; Corollary 7.2 applies to it, and the rates below are the first test of the boundary case. It has a glide-reflection pairing and, in the flat metric, a closed-form spectrum $\pi^2(m^2+n^2)$, $m+n$ even (Section 4.7). With the conformal factor $\lambda = 0.3\cos 2\pi x_1 \cos 2\pi(x_2 - \frac{1}{2})$, invariant under the glide, the spectrum has no closed form, and the reference is a Galerkin solve in the Fourier–Neumann basis $\cos(\pi m x_1)\cos(n\pi x_2)$, $\sin(\pi m x_1)\cos(n\pi x_2)$ on the double cover $[0,2]\times[0,1]$, restricted to the glide-invariant modes $m+n$ even, in which the stiffness form is diagonal and the weight is integrated by Gauss quadrature; with $|m|, n \le 30$ it is self-converged to 2×10^{-11}. The purpose is to test the paired C^r rows across a glide reflection with a nontrivial metric on an eigenvalue problem, where the multiplicities carry information: the flat band's double first eigenvalue $2\pi^2 = 19.739$ is split by the factor into 16.220 and 21.848, and the spline pencil reproduces the split. Table 7 gives the first eigenvalues and the maximum relative error over μ_1 to μ_8: the rates are 3.97, 7.88 and 11.4 for $d = 2, 4, 6$, the last from the coarsest pair, since at $n = 16$ the $(6,1)$ error 1.9×10^{-13} is at the accuracy of the reference.

8.3 Genus one: an embedded discocyte torus

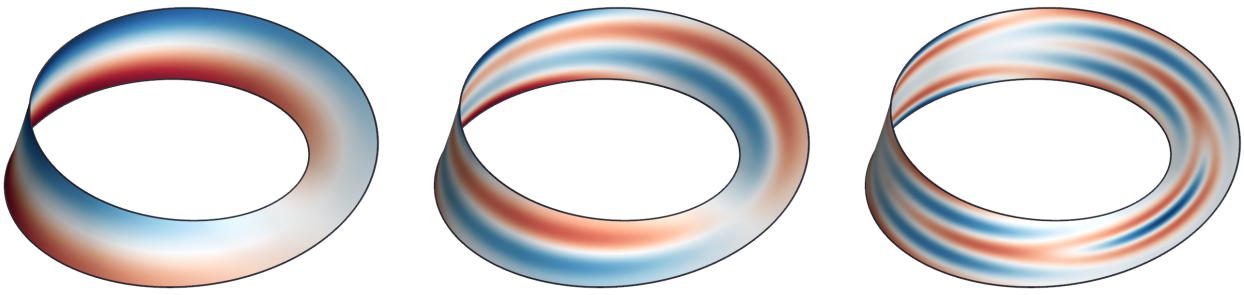
The conformal factor (19) of a torus of revolution is particularly simple, so we test the construction of Section 5.3 on a surface whose generating curve is not a circle. Let Σ be the torus of revolution generated by

$$\rho(t) = D + R\cos t, \qquad z(t) = \tfrac{1}{2}\, h(R\cos t)\, \sin t, \qquad h(r) = c_0 + c_2\Big(\frac{r}{R}\Big)^2 + c_4\Big(\frac{r}{R}\Big)^4, \qquad t \in [0, 2\pi), \tag{30}$$

revolved about an axis at distance $D = 2.4$ from the center of the profile, with $R = 1$ and the red-cell coefficients $c_0 = 0.207$, $c_2 = 2.003$, $c_4 = -1.123$ of Evans and Fung [21]. The generating curve is the meridian outline of a discocyte, the biconcave profile of the red blood cell used in the genus-zero paper [25], and Σ is that profile bent into a torus, a shape red cells occasionally

Table 7: Laplace–Beltrami eigenvalues on the Möbius band $[0,1]^2/\{(x_1,x_2)\sim(x_1+1,\,1-x_2)\}$ with free sides and the conformal metric $e^{2\lambda}g_{\mathbb{E}^2}$, $\lambda = 0.3\cos 2\pi x_1 \cos 2\pi(x_2-\frac{1}{2})$, which is invariant under the glide. Reference: Fourier–Neumann Galerkin on the double cover restricted to the glide-invariant modes, $m+n$ even, $|m|, n \le 30$, self-converged to 2×10^{-11}. The error is the maximum relative error over $\mu_1,\dots,\mu_8$; rates against $2d$. In the flat metric the first eigenvalue $2\pi^2$ is double; the conformal factor splits it into 16.2197 and 21.8482.

(d,r)	n	$\dim S_h$	μ_1	μ_2	μ_3	μ_4	max rel. error	rate ($2d$)
$(2,0)$	4	136	16.232182	21.862298	28.472943	38.275836	1.37×10^{-2}	
	8	528	16.220453	21.849319	28.367977	38.169104	1.23×10^{-3}	3.48
	16	2080	16.219725	21.848281	28.361549	38.160099	8.20×10^{-5}	3.90
	32	8256	16.219679	21.848213	28.361134	38.159515	5.22×10^{-6}	3.97
$(4,0)$	4	528	16.219681	21.848212	28.361155	38.159509	1.84×10^{-5}	
	8	2080	16.219676	21.848209	28.361106	38.159476	8.55×10^{-8}	7.75
	16	8256	16.219676	21.848209	28.361106	38.159475	3.89×10^{-10}	7.78
	32	32896	16.219676	21.848209	28.361106	38.159475	1.65×10^{-12}	7.88
$(6,1)$	4	696	16.219676	21.848209	28.361106	38.159475	4.41×10^{-8}	
	8	2704	16.219676	21.848209	28.361106	38.159475	1.66×10^{-11}	11.37
	16	10656	16.219676	21.848209	28.361106	38.159475	1.85×10^{-13}	6.49
reference			16.219676	21.848209	28.361106	38.159475		

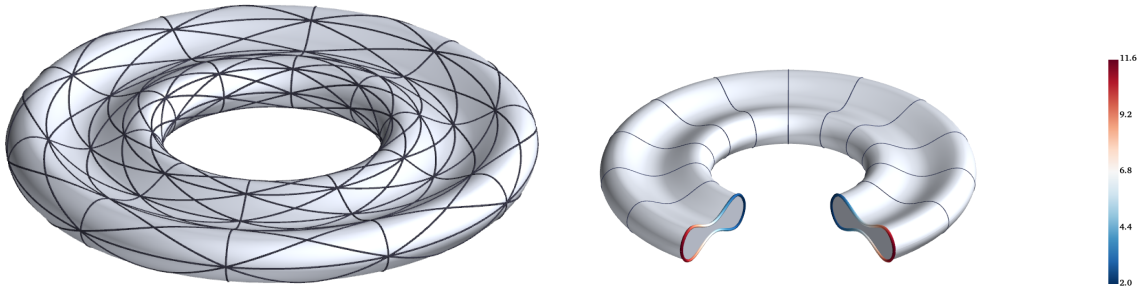


low: $\mu_1 = 16.2197$ medium: $\mu_9 = 84.1317$ high: $\mu_{25} = 272.854$

Figure 8: Eigenfunctions of the conformal Möbius band at three points of the spectrum, $d=6$, $r=1$, $n=12$, drawn on a half-twisted band in $\mathbb{R}^3$ for display; the computed metric is $e^{2\lambda}g_{\mathbb{E}^2}$ on the square, which is not the metric of the drawn band. The nodal sets refine with the eigenvalue, the boundary circle and the half twist stay visible, and the fields close up across the glide, which a failure of the paired rows would break at the seam. Common color scale; sign arbitrary.

take. The curve is regular, with speed between 0.54 and 1.05 and curvature at most 4.0. Its metric is $(\rho'^2 + z'^2)\,dt^2 + \rho^2\,d\phi^2$, and the conformal variable $u(t) = \int_0^t \sqrt{\rho'^2 + z'^2}\,/\,\rho\;ds$ makes it $\rho^2(du^2 + d\phi^2)$. Thus Σ is the flat torus $[0, U] \times [0, 2\pi]$ with $U = u(2\pi) = 2.223997463$, of modulus $\tau = 0.353960\,i$, carrying the conformal factor $w = \rho(t(u))^2$, which oscillates with $\operatorname{osc} \lambda = 0.887$ and radius ratio $\rho_{\max}/\rho_{\min} = 3.4/1.4$. The inverse $t(u)$ is computed once by quadrature and cubic spline interpolation on 2×10^4 nodes and evaluated at the quadrature points; its error is below 10^{-12}, well under the spline errors reported, and the area reproduced to ten digits below confirms it. Figure 9 shows the surface with this mesh and cut open, and Figure 10 three of its eigenfunctions.

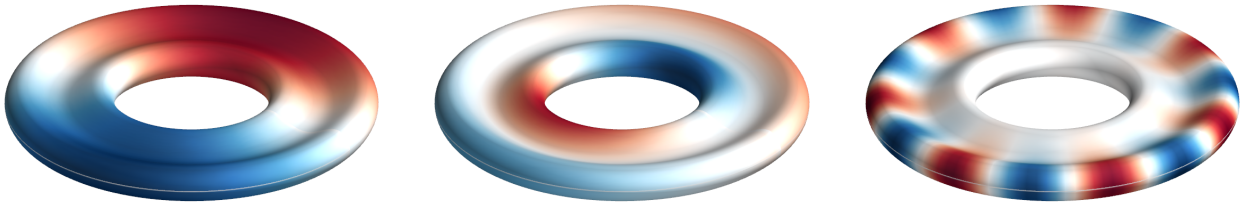
Table 8 reports the computation. The area $\int_F w\,dA$ reproduces the closed-form surface area 74.3252255767 to ten digits, a test of the conformal construction independent of the spectrum, and the eigenvalues agree to ten digits across the last three rows, which differ in degree, smoothness and mesh, so those digits are resolved. The pairs are the azimuthal modes $e^{\pm i\phi}$ exchanged by the reflection $\phi \mapsto -\phi$. Nothing in the assembly changes between this surface and the flat torus of Section 8.1: the mesh, the smoothness matrix and the stiffness matrix are the same, and the geometry enters only through w at the quadrature points.



the surface with the mesh of the flat torus carried through the conformal map

cut open at two meridians: the biconcave section, cut curves colored by w from 2.0 to 11.6

Figure 9: The discocyte torus (30) embedded in $\mathbb{R}^3$ for display. Left: the criss-cross mesh of the flat torus carried through the conformal map, its cells equal in the flat metric and spaced on the surface by the factor $w = \rho^2$. Right: the surface cut open so that the cross-section is seen. The assembly never leaves the rectangle.



$\mu_1 = 0.181376$ $\mu_{12} = 2.012077$ $\mu_{30} = 5.801418$

Figure 10: Eigenfunctions of the discocyte torus at three points of the spectrum, $d = 6$, $r = 1$ on a 12×12 mesh of the flat torus, drawn on the surface with a common color scale; the fields close up across both identifications. Sign arbitrary.

The same computation was repeated with the closed-form assembly of Section 3.4, the weight replaced by its Bernstein interpolant of degree $p = d + 2$ and no quadrature anywhere. The two assemblies agree in μ_1 to 3×10^{-9} at $(4, 0)$, $n = 16$ and 2×10^{-8} at $(6, 1)$, $n = 12$, and to 9×10^{-7} and 1.2×10^{-6} at $n = 8$, with the area behaving the same way, and the difference is the interpolation

Table 8: The discocyte torus (30): the first, third and fifth positive eigenvalues of the Laplace–Beltrami operator computed on the flat torus $[0, U] \times [0, 2\pi]$ with the conformal factor $w = \rho^2$, and the area $\int_F w\, dA$, whose closed-form value is 74.3252255767. The eigenvalues come in pairs, $\mu_1 = \mu_2$ and $\mu_3 = \mu_4$, the azimuthal modes $e^{\pm i\phi}$; μ_5 is axisymmetric. The digits stable across the last three rows, which differ in degree, smoothness and mesh, are those the computation resolves.

(d, r)	n	$\dim S_h$	area	$\mu_1 = \mu_2$	$\mu_3 = \mu_4$	μ_5
$(2, 0)$	8	512	74.3252255759	0.1814488241	0.6665909609	1.3784333938
	16	2048	74.3252255777	0.1813804583	0.6632270037	1.3496386477
$(4, 0)$	8	2048	74.3252255753	0.1813757751	0.6629779637	1.3472887073
	16	8192	74.3252255767	0.1813757738	0.6629775974	1.3472729341
$(6, 1)$	8	2624	74.3252255759	0.1813757738	0.6629775961	1.3472728660
	12	5904	74.3252255767	0.1813757738	0.6629775958	1.3472728607
$(10, 1)$	6	5220	74.3252255767	0.1813757738	0.6629775958	1.3472728607

error of the weight, which has large derivatives near the dimples of the profile; it enters as the δ of Proposition 6.2(iii) and decreases with h. The closed-form assembly is the one for which the flat theory of Section 6 holds with nothing observed; the Gauss-rule assembly is the more accurate of the two at coarse meshes, and the tables above report it.

8.4 Genus one: comparison with surface finite elements at equal degree

The surface is embedded, so the parametric surface finite element method applies to it, and the comparison can be made at equal polynomial degree, which parity forbids on the curved models. For $k = 2, 4, 6$ Lagrange elements of degree k are assembled on the same parameter mesh in two ways: on the polyhedral surface whose vertices lie on Σ, with the geometry interpolated at degree one, and isoparametrically, with the geometry interpolated at degree k from nodes on Σ [19, 16]. Against them stands the conformal space S_k^0 of Table 8, which on a triangle mesh is the same Lagrange space P_k; the only difference between the three columns of Table 9 is therefore how the geometry enters, and the purpose of the experiment is to measure that difference alone. The reference is $\mu_1 = 0.1813757738$, on which the last three rows of Table 8 agree, and the area is compared with its closed-form value.

Three things are read from the table. On flat facets the error is the geometric error of order h^2, and raising the degree from 2 to 6 changes neither the area nor the eigenvalue beyond the fourth digit: at 96×48 all three degrees give 3.3×10^{-3} in area and 2.9×10^{-3} in μ_1. Isoparametric elements recover the order of the geometry, reaching 1.9×10^{-9} in μ_1 at $k = 4$ on 73728 unknowns and 1.5×10^{-9} at $k = 6$ on 41472. The conformal splines reach the same accuracy with fewer unknowns because no geometric error is present: S_4^0 is below 10^{-10} on 8192 unknowns, and S_6^1 on 2624. This is the comparison the analysis of Section 7 predicts, and it is made at equal degree and on the same mesh.

8.5 Genus two: the Bolza surface

The Bolza surface is the quotient of $\mathbb{H}^2$ by the group generated by the opposite-side pairings of the regular octagon with angles $\pi/4$ (Figure 5), a closed orientable two-manifold of genus two, a sphere with two handles, carrying a hyperbolic metric of area 4π. It is the genus-two surface whose spectrum is known to high precision: Strohmaier and Uski [39] give $\mu_1 = 3.8388872588421995$ with multiplicity three. The experiment tests the two layers where the hyperbolic case differs from the

Table 9: Equal-degree comparison on the discocyte torus (30). For $k = 2, 4, 6$, Lagrange elements of degree k are run on the polyhedral surface interpolating the parameter mesh (geometry of degree 1) and isoparametrically (geometry of degree k), and the conformal spline space S^0_k on the flat torus with the factor w, which on a triangle mesh is the same space P_k with the geometry carried by w. Errors are relative to $\mu_1 = 0.1813757738$, on which the last three rows of Table 8 agree, and to the closed-form area 74.3252255767. The flat-facet rows are limited to the geometric error of order h^2 whatever the degree; the isoparametric rows improve with the degree of the geometry; the conformal rows carry no geometric error.

k	method	unknowns	area rel. error	μ_1 rel. error
2	P_2, flat facets, 24×12	1164	5.1×10^{-2}	4.8×10^{-2}
	P_2, flat facets, 48×24	4608	1.3×10^{-2}	1.2×10^{-2}
	P_2, flat facets, 96×48	18432	3.3×10^{-3}	2.9×10^{-3}
	P_2, isoparametric, 24×12	1164	6.1×10^{-4}	5.2×10^{-4}
	P_2, isoparametric, 48×24	4608	4.3×10^{-5}	3.5×10^{-5}
	P_2, isoparametric, 96×48	18432	2.7×10^{-6}	2.2×10^{-6}
	S^0_2 conformal, $n = 8$	512	1.1×10^{-11}	4.0×10^{-4}
	S^0_2 conformal, $n = 16$	2048	1.3×10^{-11}	2.6×10^{-5}
4	P_4, flat facets, 24×12	4620	5.1×10^{-2}	4.6×10^{-2}
	P_4, flat facets, 48×24	18432	1.3×10^{-2}	1.2×10^{-2}
	P_4, flat facets, 96×48	73728	3.3×10^{-3}	2.9×10^{-3}
	P_4, isoparametric, 24×12	4620	9.5×10^{-6}	6.8×10^{-6}
	P_4, isoparametric, 48×24	18432	2.4×10^{-7}	1.3×10^{-7}
	P_4, isoparametric, 96×48	73728	3.6×10^{-9}	1.9×10^{-9}
	S^0_4 conformal, $n = 8$	2048	1.9×10^{-11}	6.9×10^{-9}
	S^0_4 conformal, $n = 16$	8192	1.8×10^{-13}	6.0×10^{-11}
6	P_6, flat facets, 24×12	10380	5.1×10^{-2}	4.6×10^{-2}
	P_6, flat facets, 48×24	41472	1.3×10^{-2}	1.2×10^{-2}
	P_6, isoparametric, 24×12	10380	3.1×10^{-7}	2.5×10^{-7}
	P_6, isoparametric, 48×24	41472	2.8×10^{-9}	1.5×10^{-9}
	S^1_6 conformal, $n = 8$	2624	1.1×10^{-11}	1.3×10^{-10}
	S^1_6 conformal, $n = 12$	5904	6.6×10^{-13}	$< 10^{-10}$

flat one, the corner-cycle closure of the pairings and the Klein-chart quadrature of the area density, against an external reference.

Table 10 reports the computation on S^0_d at three refinement levels. Three things are read from it. The area defect is at roundoff at every level, so the polygon, its identifications and the quadrature reproduce $-2\pi\chi = 4\pi$; this is the third rung of Section 4.6 measured on the meshes actually used. The computed $\mu_{0,h}$ is zero to 10^{-13}, which is Proposition 6.2(ii) and requires the constants to lie in $\mathcal{B}_d$, hence the even degrees. And the relative error of μ_1 falls at the rates 3.85, 7.82 and 11.84 for $d = 2, 4, 6$ against the predicted $2d$ of Corollary 7.2, reaching 4.5×10^{-12} at $d = 6$ on 9214 degrees of freedom. The computed eigenvalues are above the reference at every level, as Proposition 6.2(i) requires, and the first cluster has multiplicity three, the three computed values agreeing to 1.4×10^{-7} at $d = 6$, $L = 2$.

Table 10: The Bolza surface with its hyperbolic metric: first nonzero eigenvalue on $S^0_d(\mathbb{H}^2/\Gamma, \triangle)$ at three refinement levels, against the high-precision value $\mu_1 = 3.8388872588421995$ of Strohmaier and Uski [39]. The area column is the defect of $\int_F dA_{\mathbb{H}^2}$ against $4\pi = -2\pi\chi$ under the assembly quadrature, the zero column is the computed μ_0, which is reproduced to roundoff because the constants lie in $\mathcal{B}_d$ at even degree, and the rates are measured against the predicted $2d$ of Corollary 7.2.

d	L	triangles	$\dim S^0_d$	area defect	$\mu_{0,h}$	$\mu_{1,h}$	rel. error (rate)
2	1	32	62	1.42×10^{-14}	4.44×10^{-16}	4.307173488000	1.22×10^{-1}
	2	128	254	1.78×10^{-15}	2.66×10^{-15}	3.873155718099	8.93×10^{-3} (3.77)
	3	512	1022	5.33×10^{-15}	1.78×10^{-15}	3.841264437198	6.19×10^{-4} (3.85)
4	1	32	254	1.42×10^{-14}	9.99×10^{-15}	3.852134233186	3.45×10^{-3}
	2	128	1022	1.78×10^{-15}	2.66×10^{-14}	3.838936644476	1.29×10^{-5} (8.07)
	3	512	4094	7.11×10^{-15}	4.80×10^{-14}	3.838887477795	5.70×10^{-8} (7.82)
6	1	32	574	1.95×10^{-14}	6.59×10^{-14}	3.839143822147	6.68×10^{-5}
	2	128	2302	1.78×10^{-15}	1.09×10^{-13}	3.838887322669	1.66×10^{-8} (11.97)
	3	512	9214	0	1.38×10^{-13}	3.838887258860	4.54×10^{-12} (11.84)

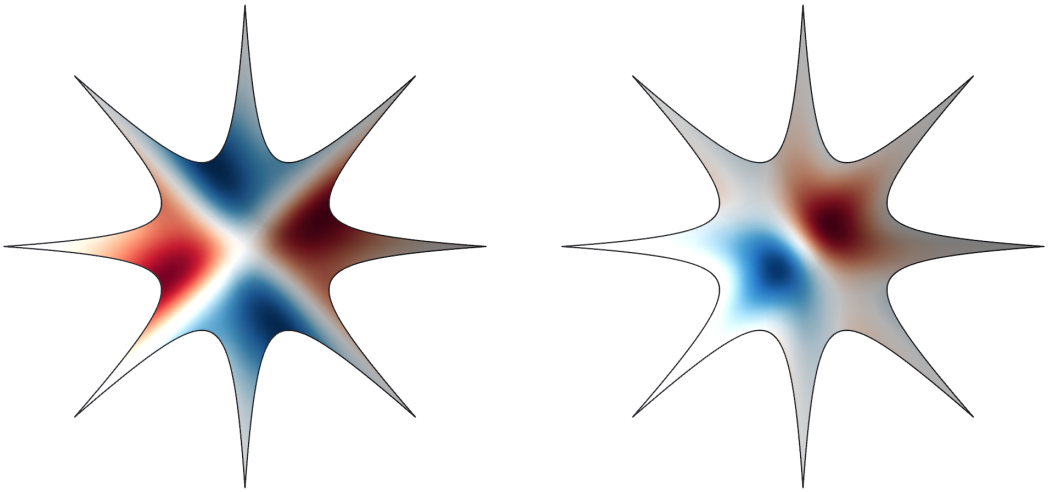
The whole spectrum, not only its first value, tests the identifications, because the multiplicities are forced by the automorphism group of the surface and are destroyed by a wrong pairing or a corner walk composed in the wrong order. Table 11 compares the first ten positive eigenvalues computed at $d = 6$, level 2, with the high-precision values of [39]. All ten are reproduced, with relative errors from 1.7×10^{-8} at λ_1 to 2.9×10^{-6} at λ_{10}, and the multiplicities come out as $3, 4, 2, 4, 3, 3, 4, 1, 3, 4$, which are the values established in [22, 15]; the clusters split by at most 8×10^{-5}, four orders below the gaps between consecutive distinct eigenvalues, so the multiplicities are unambiguous. Figure 11 shows one eigenfunction from each of the first two eigenspaces, drawn on the octagon.

8.6 Increasing genus: a ladder from $g = 2$ to $g = 8$

Nothing in the construction changes with the genus except the polygon and its pairings; this experiment tests that claim. The regular $4g$-gon with interior angles $2\pi/4g$ and opposite sides paired presents a closed orientable surface of genus g for every $g \geq 2$: the corner walk $j \mapsto j + 2g + 1$ has a single cycle since $\gcd(2g + 1, 4g) = 1$, the angle sum is $4g \cdot 2\pi/4g = 2\pi$, and the area is $4\pi(g - 1)$; at $g = 2$ this is the Bolza octagon. Table 12 runs the same code, with the polygon as the only input, for $g = 2, \ldots, 8$ on S^0_4, and reports the area defect, the computed zero mode and the first eigenvalues. The computed cluster at the first positive eigenvalue contains three values at $g = 2$, the Bolza multiplicity, and at least two at every $g \geq 3$, the second and third computed values agreeing to the digits shown; the full multiplicities above $g = 2$ were not resolved, since only five eigenvalues

Table 11: The first ten positive eigenvalues of the Bolza surface computed on $S_6^0(\mathbb{H}^2/\Gamma, \triangle)$ at level 2 (2302 degrees of freedom), against the high-precision values of Strohmaier and Uski [39], truncated here to twelve digits. The multiplicity column is the number of computed eigenvalues in the cluster and the spread is its width; the multiplicities $3, 4, 2, 4, 3, 3, 4, 1, 3, 4$ are those established in [22, 15] and are forced by the automorphism group of the Bolza surface, which has order 96, the largest in genus two. A wrong side pairing or a corner walk composed in the wrong order changes them.

j	λ_j computed	λ_j reference	mult.	spread	rel. error
1	3.838887344	3.838887258842	3	6×10^{-8}	2.2×10^{-8}
2	5.353601449	5.353601341189	4	7×10^{-8}	2.0×10^{-8}
3	8.249555650	8.249554815201	2	8×10^{-7}	1.0×10^{-7}
4	14.726219099	14.726216787789	4	2×10^{-6}	1.6×10^{-7}
5	15.048922049	15.048916133267	3	5×10^{-6}	3.9×10^{-7}
6	18.658834379	18.658819627260	3	7×10^{-6}	7.9×10^{-7}
7	20.519866153	20.519859734142	4	3×10^{-7}	3.1×10^{-7}
8	23.078590433	23.078558481382	1	0	1.4×10^{-6}
9	28.079666593	28.079605737678	3	6×10^{-6}	2.2×10^{-6}
10	30.833132698	30.833042737933	4	8×10^{-5}	2.9×10^{-6}



$\mu_1 = 3.8389$, multiplicity 3 $\mu_4 = 5.3537$, multiplicity 4

Figure 11: Eigenfunctions of the Bolza surface drawn on the fundamental octagon on the hyperboloid, viewed along the x_0 axis; one member of each of the first two eigenspaces. Sign and scale are arbitrary.

were computed per configuration. The first positive eigenvalue decreases with g, from 3.839 at $g = 2$ to 0.469 at $g = 8$, consistent with the bound $\mu_1 \le 8\pi(g+1)/\operatorname{Area} = 2(g+1)/(g-1)$ of Yang and Yau [42] and with μ_1 being small on surfaces with long thin parts. No reference values exist for these surfaces above $g = 2$; the table stands on its own consistency, the area defect and the zero mode being the checks, and the values at level 2 are the ones to compare against.

Table 12: Genus ladder: the regular $4g$-gon with opposite sides paired, $g = 2, \dots, 8$, on S_4^0 at levels 1 and 2. For every g the corner walk $j \mapsto j + 2g + 1 \pmod{4g}$ visits all $4g$ corners in one cycle ($\gcd(2g+1, 4g) = 1$), the angle sum is 2π, and the quotient is a closed orientable surface of genus g with hyperbolic area $4\pi(g-1)$; the area column is the defect of the assembly quadrature against that value. The columns μ_1 and μ_3 give the first and third computed positive eigenvalues; their agreement at $g \ge 3$ shows a cluster of multiplicity at least two, and at $g = 2$ the cluster is the Bolza triple. At $g = 8$ only level 1 is available; see the text.

g	n	χ	level	$\dim S_4^0$	area defect	$\mu_{0,h}$	μ_1	μ_3
2	8	-2	1	254	8.1×10^{-13}	0.0×10^{0}	3.852134	3.852532
			2	1022	3.6×10^{-15}	1.3×10^{-14}	3.838937	3.838937
3	12	-4	1	380	6.2×10^{-8}	3.3×10^{-15}	1.830331	1.830331
			2	1532	5.7×10^{-12}	3.8×10^{-15}	1.807001	1.807001
4	16	-6	1	506	7.4×10^{-6}	3.6×10^{-15}	1.173504	1.173504
			2	2042	5.0×10^{-10}	1.1×10^{-15}	1.142510	1.142510
5	20	-8	1	632	9.9×10^{-5}	7.1×10^{-15}	0.857188	0.857188
			2	2552	8.6×10^{-9}	2.9×10^{-15}	0.820911	0.820911
6	24	-10	1	758	4.7×10^{-4}	1.0×10^{-14}	0.673217	0.673217
			2	3062	6.2×10^{-8}	1.0×10^{-14}	0.634085	0.634085
7	28	-12	1	884	1.1×10^{-3}	6.4×10^{-15}	0.553080	0.553080
			2	3572	2.7×10^{-7}	1.3×10^{-15}	0.513179	0.513179
8	32	-14	1	1010	1.2×10^{-3}	1.2×10^{-14}	0.468515	0.468515

The ladder also exposes the limitations of the apparatus, which we state because they are the practical content of this experiment.

(i) *The regular polygon is a poor fundamental domain at large genus.* Its corner angles are $2\pi/4g$, so the coarse triangles have angles below 12° at $g = 4$ and below 6° at $g = 8$, and the shape-regularity constant in Theorem 7.1 degrades accordingly. This shows in the area defect at level 1, which grows from 10^{-8} at $g = 3$ to 10^{-3} at $g = 8$, and in the gap between the level-1 and level-2 eigenvalues, which grows from 1% at $g = 3$ to 8% at $g = 7$. The pants decomposition of Section 4.4 is the remedy, with all angles $\pi/2$ at every genus; the construction does not depend on the polygon being regular, and only the mesh layer changes.

(ii) *The circumradius grows with g and the hyperboloid coordinates with it.* The corners of the regular $4g$-gon lie at hyperbolic distance R from the center with $\cosh R = \cot^2(\pi/4g)$, the regular n-gon with interior angle α having $\cosh R = \cot(\pi/n)\cot(\alpha/2)$ with $n = 4g$ and $\alpha = \pi/2g$; so the hyperboloid coordinate $x_0 = \cosh R$ is 5.8 at $g = 2$, 103 at $g = 8$, and grows like $16g^2/\pi^2$. The identification of paired boundary points is done with an absolute tolerance, and at $g = 8$, level 2, the roundoff in the transported vertices reaches 10^{-7} and the match fails; the same happens at level 1 for $g \ge 9$. This is an implementation limit of the single-polygon presentation, cured by a relative tolerance or by the pants decomposition, whose pieces each stay near the origin of their own chart, and it is the reason the table stops where it does.

(iii) *The Klein-chart density steepens and the conformal class is fixed by the polygon.* The corners approach the unit circle as g grows, so the coarse triangles need a quadrature order that grows

with g; and the regular $4g$-gon is one point in a moduli space of dimension $6g-6$, so the ladder tests the machinery across genus, not the dependence of the spectrum on the conformal class.

(iv) *No external reference above genus two.* The Bolza spectrum is the only published high-precision reference in this family; above it the checks are internal, the area defect and the zero mode.

9 Discussion and conclusions

Write a closed two-manifold as X/Γ with a conformal metric $e^{2\lambda}g_X$, where X is the round sphere with Γ trivial in genus zero, the flat plane with Γ a discrete group of Euclidean motions when $\chi = 0$, and the hyperbolic plane with Γ a discrete group of hyperbolic isometries when $\chi < 0$. Lemma 5.1 puts the area in the weight and leaves the Dirichlet energy in the model, so in all three cases the stiffness matrix is the Dirichlet matrix of the model on the fundamental domain and the geometry is one scalar at the quadrature points. The identifications enter as rows of the smoothness matrix and nowhere else (Proposition 4.1), the same rows serving orientation-preserving and orientation-reversing pairings, and by Section 4.4 the polygon itself is inessential: a triangulation with an isometry per edge is enough. The genus therefore changes the fundamental domain and its gluing, and neither the spline spaces nor the assembly.

The numerical analysis is that of a conforming method on a domain represented without approximation in each of the three geometries. Corollary 7.2 carries no geometric consistency term, the only variational crime is quadrature, of the mass form alone on the flat quotients and of both forms on the hyperbolic ones, and a perturbation of the weight itself acts as a relative error uniform along the spectrum (Proposition 6.2 and the remark following it); for a fixed quotient the conformal factor enters the constants only through $w_{\min}$, $w_{\max}$ and $\|w\|_{C^1}$. The experiments bear this out, the same rates appearing on the torus, the Klein bottle, the Möbius band and the Bolza surface. The one model-dependent feature is parity: constants are representable at every degree in the plane and only at even degree on the sphere and the hyperboloid. With the locally supported null-space basis of Section 3.2, the reduced operator scales like h^{-2} and the reduced mass matrix is conditioned uniformly in h, with a level-independent stencil (Section 7.1); the constants of that scaling depend on the scaling of the basis and are not compared here with an unconstrained one.

Three limitations are stated. The uniformizing data, the fundamental domain with its gluing and the conformal factor, are input; for a surface given as a mesh they come from a discrete uniformization with its own error. The conformal class enters through finitely many parameters that sit inside the gluing rather than in the weight, so a sweep within a class is cheap and a change of class rebuilds the domain. And a factor with strong conformal concentration under-resolves on a quasi-uniform mesh; the experiments use quasi-uniform meshes throughout.

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